



Time-dependent vehicle routing problem with path flexibility



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ABSTRACT

Conventionally, vehicle routing problems are defined on a network in which the customer locations and arcs are given. Typically, these arcs somehow represent the distances or expected travel time derived from the underlying road network. When executed, the quality of the solutions obtained from the vehicle routing problem depends largely on the quality of the road network representation. This paper explicitly considers path selection in the road network as an integrated decision in the time-dependent vehicle routing problem, denoted as path flexibility (PF). This means that any arc between two customer nodes has multiple corresponding paths in the road network (geographical graph). Hence, the decisions to make are involving not only the routing decision but also the path selection decision depending upon the departure time at the customers and the congestion levels in the relevant road network. The corresponding routing problem is a time-dependent vehicle routing problem with path flexibility (TDVRP-PF). We formulate the TDVRP-PF models under deterministic and stochastic traffic conditions. We derive important insights, relationships, and solution structures. Based on a representative testbed of instances (inspired on the road network of Beijing), significant savings are obtained in terms of cost and fuel consumption, by explicitly considering path flexibility. Having both path flexibility and time-dependent travel time seems to be a good representation of a wide range of stochasticity and dynamics in the travel time, and path flexibility serves as a natural recourse under stochastic conditions. Exploiting this observation, we employ a Route-Path approximation method generating near-optimal solutions for the TDVRP-PF under stochastic traffic conditions.

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1. Introduction

Urbanization has become a worldwide phenomenon during the past decades. Around 54% of the world's population live in cities in 2014 and this is expected to be 66% by 2050. Although in the most urbanized region, i.e., Northern America, 82% of the population live in urban area, only 40% and 48% of the population live in urban areas in Africa and Asia, respectively (United Nations, 2015). However, the rates of urbanization in Africa and Asia are faster than those of other regions, which results in more and more emerging mega-cities in these two regions, such as China and India.

Hand in hand with urbanization (Savelsbergh and Van Woensel, 2016), traffic congestion becomes one of the major challenges not only for commuters but also for logistics companies. For example, Beijing is ranked 15 among over 200 large

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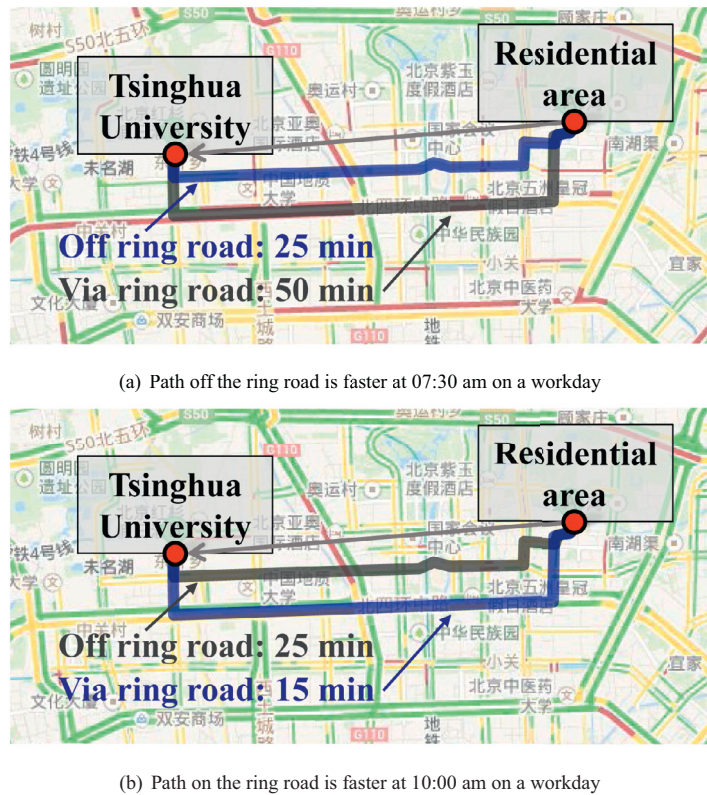


Fig. 1. Two popular morning commute paths from a residential area to Tsinghua University in Beijing, China.

cities (population > 800K) in the world, based on the overall congestion level in 2014 (TomTom, 2015). Congestion significantly increases the fuel consumption of vehicles, and therefore increases the cost of logistics companies. In 2011, the cost of delays on roads in the six largest capital cities in Australia is estimated to be around \$13.7 billion (Infrastructure Australia, 2015). In China, fuel cost accounts for over 46% of the total operational cost for a logistics company (China Federation of Logistics & Purchasing, 2014).

First mile pickups and last mile deliveries, often modeled as a vehicle routing problem, are more and more situated in these congested urbanized regions. Vehicle routing problems are mostly defined on a customer-based graph. Arcs between these customers are assumed to adequately represent the distances or expected travel time. As such, the path used by a vehicle (in the road network) between a pair of customers is aggregated into the arc cost. When executed, the quality of the solutions obtained depends largely on the quality of this road network representation. Typically, in the literature (see e.g., Fleischmann et al., 2004; Gendreau et al., 2015; Ichoua et al., 2003; Jabali et al., 2012), this traffic dependency is modeled implicitly, resulting in time-dependent and/or stochastic travel time variants of the vehicle routing problem.

This paper takes an alternative approach and explicitly considers path selection in the road network as an integrated decision in the vehicle routing problem that minimizes the total cost, including fuel cost and vehicle depreciation cost. This means that any arc between two customer nodes has multiple corresponding paths in the road network (geographical graph). Hence, the decisions to make are involving not only the routing decision but also the path selection decision depending upon the departure time at the customers and the congestion levels in the relevant road network. We denote the path selection decision as path flexibility (PF). The corresponding routing problem is a time-dependent vehicle routing problem with path flexibility (TDVRP-PF). We formulate the TDVRP-PF models under deterministic and stochastic traffic conditions.

Consider the example given in Fig. 1, there are two typical path options, one via the 4th Ring Road (the lower path) and the other off the ring road (the upper path). Between these two paths, the faster path depends on the actual departure time and on the (spread of the) traffic congestion in the road network. Note that the congestion in the road network is also region-dependent. Consider the example in Fig. 2, where the colors of the roads represent the traffic conditions (green for free flow, yellow for medium congestion, and red for heavy congestion). Observe that the traffic conditions of the ring roads change significantly between peak and off-peak hours, while those of the roads within the 2nd ring remain almost unchanged. Jointly considering routing and path selection decisions makes it possible to explicitly consider the temporal and spatial differences of congestion in the road network.

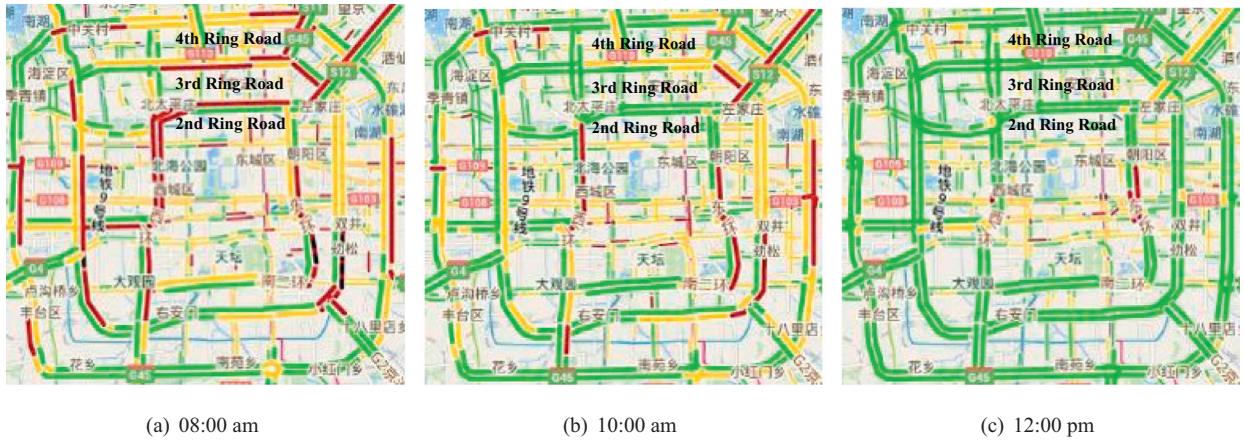


Fig. 2. Estimated traffic conditions of urban area in Beijing on Mondays (Source: Google Map).

There have been few studies explicitly considering path flexibility in the vehicle routing problem. [Garaix et al. \(2010\)](#) worked on a multi-attribute vehicle routing problem and proposed a multigraph to represent the network. [Setak et al. \(2015\)](#) solved the time-minimizing time-dependent vehicle routing problem in a multigraph with First-In-First-Out (FIFO) property via tabu search. [Ehmke et al. \(2016b\)](#) and [Qian and Eglese \(2016\)](#) both dealt with the time-dependent vehicle routing problem to minimize emissions, which align with our work well. Our paper fits into this stream of literature. We focus on the analysis of optimization models and their optimal solutions, while [Ehmke et al. \(2016b\)](#) and [Qian and Eglese \(2016\)](#) proposed efficient and effective tabu search methods to solve larger instances. We flexibly and explicitly keep the precomputed paths as candidate paths and the path selection is embedded as decision variables in the TDVRP-PF, under deterministic and stochastic traffic conditions.

Our paper makes a number of contributions to the literature.

1. We explicitly consider path flexibility in the time-dependent vehicle routing problem (TDVRP-PF). We model the TDVRP-PF under deterministic and stochastic traffic conditions, the latter of which leads to a two-stage stochastic program. Rather than developing advanced solution algorithms for our problem on-hand, we use an industrial solver (CPLEX) and derive important insights, relationships, and structures. We observe that, when the TDVRP and TDVRP-PF solutions are structurally different, the cost saving of path flexibility is usually more significant. This is important in practice. For instance, home appliance delivery companies load the appliances (normally big and heavy) according to their delivery sequence. It is difficult to change the sequence en route. These results serve as a good basis for further algorithmic work, which is left for future research.
2. We demonstrate the benefits of path flexibility based on a representative testbed of instances inspired on the road network of Beijing (i.e., similar to [Qian and Eglese, 2016](#), who used the London road network for their instances). For this rich set of instances, significant savings are obtained in terms of cost and fuel consumption by explicitly considering path flexibility into the routing and scheduling. It is also demonstrated that these improvements are more significant than time flexibility (i.e., flexible departure time at the depot and/or post-service waiting time at customer sites). Additionally, these observations are robust based on our extensive sensitivity analysis, with regard to the customer network topology and the corresponding choices on the number of alternative paths in the geographical graph.
3. We observe that the routes generated with the TDVRP-PF under deterministic traffic conditions are very good approximations for the TDVRP-PF under stochastic traffic conditions. Having both path flexibility and time-dependent travel time seems to be a good representation of a wide range of stochasticity and dynamics in the travel time, and path flexibility serves as a natural recourse under stochastic conditions. Exploiting this observation, we employ a Route-Path approximation method generating near-optimal solutions for the TDVRP-PF under stochastic traffic conditions.

In the remainder of this paper, we first review the relevant literature in [Section 2](#), and then formally describe the TDVRP-PF in [Section 3](#). In [Section 4](#), we provide some preliminaries on fuel consumption and time-dependent models, and we formulate the TDVRP-PF as mixed-integer programs in [Section 5](#), including both deterministic and stochastic versions. We describe the design of experiments and discuss the numerical results in [Section 6](#). Finally, we summarize our work and provide the suggestions for future research in [Section 7](#).

2. Literature review

Our work is closely related to the green vehicle routing problem (green VRP), which considers carbon emission or fuel consumption of vehicles, and the time-dependent vehicle routing problem (TDVRP), which reflects the varying traffic conditions. Interested readers can refer to [Lin et al. \(2014\)](#) and [Demir et al. \(2014b\)](#) for reviews of the green VRP and

Gendreau et al. (2015) for a recent review of the TDVRP. As shortly outlined in the introduction, our paper extends and builds upon the literature in a number of directions.

The conventional time-minimizing TDVRP provides the foundation of our research on the time-dependency modelling. It is standard to use time-dependent time matrices to store the fastest paths of different departure time. Due to the First-In-First-Out (FIFO) network, it is optimal to always choose the fastest path corresponding to a certain departure time. However, this method cannot be extended to a cost-minimizing TDVRP, where the challenge is, as briefly mentioned in Savelsbergh and Van Woensel (2016), the costs are dependent on speed, distance, and vehicle load. Due to the complex structure, it may not be globally optimal for the routing solutions to choose the locally cost-minimizing path between each pair of customers in a route. Alternatively, we keep multiple paths as the pool of candidate paths and propose a corresponding mathematical model allowing the simultaneous decisions of both routing and path selection. In their paper, Koç et al. (2016) analyzed the impact of spatial differences of congestion (speed zones). We focus on path flexibility considering both spatial and temporal differences. Ehmke et al. (2016b) and Qian and Eglese (2016) proposed metaheuristic methods, while we focus on obtaining insights and the analysis of optimal solutions.

Although our work is similar to the Pollution-Routing Problem (PRP, Bektaş and Laporte, 2011), there are several major differences in the problem settings. First, we do not consider the drivers' salary. Second, we do not control the vehicle speed as we focus on the decisions of routing and path selection, which is usually decided by the planning department, rather than the detailed driving behavior. Further, as the roads are usually congested, we assume that the vehicles just follow traffic. Last, we consider the routing problem in a time-dependent traffic condition and explicitly incorporate decisions of path selection in the routing problem.

2.1. Time-dependent routing

The TDVRP has been introduced to capture congestion in a traffic network. Malandraki and Daskin (1992) proposed a mixed integer formulation of the time-dependent vehicle routing problem where the travel time on each arc is a step function of the departure time. Malandraki and Dial (1996) applied restricted dynamic programming to solve the time-dependent traveling salesman problem. Hill and Benton (1992) introduced a model for time-dependent travel speeds and methods for parameter estimation, which can be applied for the time-dependent vehicle routing problems. However, all the studies disrespect the FIFO property in the time-dependent network, that is, vehicles leaving later may arrive earlier on the same arc.

Two methods (Fleischmann et al., 2004 and Ichoua et al., 2003) appeared to model the time-dependent network that respects the FIFO property. Fleischmann et al. (2004) modeled the travel time via the smoothed travel time function based on the step function of travel time. Ichoua et al. (2003) presented a model where travel speeds are step functions and travel time become piecewise linear functions. These methods are commonly applied in research afterwards.

The underlying time-dependent shortest path problem (TDSPP) is essential to the TDVRP. Fleischmann et al. (2004) anticipated the potential value of considering path changing in the TDVRP and suggested it as the research direction. Eglese et al. (2006) constructed a timetable of time-dependent shortest paths based on the real-world road network and the corresponding floating vehicle data in England. Kok et al. (2012) solved the TDVRP with time-dependent shortest paths via a restricted dynamic programming heuristic and showed the significant reduction in travel time to use the time-dependent shortest paths in the TDVRP. Similarly, Ehmke et al. (2012) also showed the superiority of time-dependent vehicle routing over static vehicle routing in a city logistics case study. They also provided detailed analyses of spatio-temporal structures of the time-dependent routing solutions.

Maden et al. (2010) and Figliozzi (2011) derived the daily speed levels based on the speed observed in 15-minute time bins averaged over 3 month period of real traffic data, in order to provide a good approximation. Alternatively, studies such as Kuo (2010), Jabali et al. (2012), and Franceschetti et al. (2013) use fewer (from 2 to 5) speed levels to depict typical traffic conditions in urban areas. Following this trend, we study the situation of 3 speed levels in the morning shift, where the second speed level is of the peak hour. Although Ehmke et al. (2016b) and Qian and Eglese (2016) use finer speed levels to depict the 24 hour traffic conditions, the delivery time horizon in their instances usually contain only one peak hour period.

2.2. Green routing

There have been increasing and various studies concentrating on the green vehicle routing problem in recent years. Erdoğan and Miller-Hooks (2012) considered a fleet of alternative fuel-powered vehicles with the restriction of limited driving range and limited refueling infrastructure, and thus the necessity to include the decision of visiting fuel station in the vehicle routing problem. Kopfer and Kopfer (2013) and Kwon et al. (2013) considered the effect of a heterogeneous fleet of vehicles in the green vehicle routing problem. Research on the case study of real world practice includes Ubeda et al. (2011), Figliozzi (2011), etc.

Pollution-Routing Problem is an important variant of the green VRP. Bektaş and Laporte (2011) first coined the Pollution-Routing Problem (PRP), which also first used the Comprehensive Modal Emission Model (CMEM) to calculate fuel consumption. Demir et al. (2014a) extended the PRP to a bi-objective PRP where the objective functions are to minimize both fuel consumption and driving time. Koç et al. (2014) considered the PRP with a heterogeneous fleet of vehicles, to determine the number of vehicles of each vehicle type and the routing of these vehicles, in order to minimize the routing cost and

the fixed cost. Tajik et al. (2014) dealt with the PRP with pickups and deliveries under uncertainty and used the robust optimization approach to solve the problem. Koç et al. (2016) considered emission minimization within the location-routing problem with a heterogeneous fleet, where the city is divided into different zones, each with a constant (time-independent) speed. As the vehicle travels among the zones with different load, it may choose different paths between zones to minimize emissions.

On the perspective of algorithmic design, Demir et al. (2012) proposed an adaptive large neighborhood search algorithm to solve the PRP. Kramer et al. (2015b) developed a hybrid iterated local search approach with a set partitioning procedure and a speed optimization algorithm. Kramer et al. (2015a) further proposed an exact method on speed and departure time optimization, which improves the performance of the iterated local search method developed by Kramer et al. (2015b). Dabia et al. (2016) presented an exact method based on a branch-and-price algorithm, including new dominance criteria to discard unpromising labels.

2.3. Green time-dependent routing

As congestion significantly affects fuel consumption, there are a stream of studies taking fuel consumption or carbon emission into consideration in the time-dependent vehicle routing problems, which we refer to as the Green TDVRP.

Maden et al. (2010) proposed a LANTIME heuristic algorithm to minimize the total travel time. They calculated the emissions of the time-minimizing solutions and showed a 7% emission reduction in the case study. Kuo (2010) introduced the TDVRP to minimize fuel consumption and solved the problem using simulated annealing. The result showed a clear trade-off between time-minimizing and fuel-minimizing solutions. Figliozzi (2011) minimized fuel consumption in the TDVRP in urban freight distribution based on the real network of Portland. Jabali et al. (2012) studied a time-dependent green vehicle routing problem, which uses the Methodology for Calculating Transportation Emissions and Energy Consumption (MEET, Hickman et al., 1999) emission model to evaluate the CO₂ emissions. However, these studies ignore the load when calculating the emissions.

Franceschetti et al. (2013) introduced the time-dependent pollution-routing problem, using a mixed integer programming formulation. They obtained the analytical results of speed and departure time optimization based on a two-step speed function with congestion first and free flow second, where travel speeds in the free flow are decisions. However, their results are difficult to generalize to speed functions with more than two steps. Soysal et al. (2015) considered a time-dependent two-echelon vehicle routing problem, which minimizes the total cost including fuel cost for the two echelons, drivers' cost for the two echelons, and handling cost in the satellites (second-echelon depots). The numerical experiment also uses a two-step speed function with congestion first and free flow second. Both studies depict time-dependency as a two-step speed function on each arc without considering the underlying path(s). Therefore, the corresponding travel time function and fuel consumption function on each arc are predetermined before routing. When the underlying paths are considered, as in our work, the travel time function and fuel consumption function associated with each arc depend on the selected path and vehicle load, where the latter is determined by the routing decisions.

The time-dependent emission-minimizing path problem, the underlying problem of the green TDVRP, is somehow understudied. Unlike the TDSP, it is generally very difficult or impossible to calculate the time-dependent emission-minimizing path, as it requires the information of both departure time and vehicle load on each arc, which depend on the routing decision. It might be possible to precompute the paths based on every combination of discretized departure time and vehicle load, which is by itself computationally demanding and thus makes the routing problem overcomplicated. Therefore, the existing studies (e.g., Qian and Eglese, 2014 and Ehmke et al., 2016a) use emission models that neglect the vehicle load, such as MEET and National Atmospheric Emission Inventory (NAEI, 2013).

Ehmke et al. (2016b) dealt with the time-dependent vehicle routing problem to minimize emissions. They investigated how to simplify the precomputation of emission-minimizing paths via identifying the optimal paths that are independent of the vehicle load and used a tabu search method to solve the routing problem. Their test instances are based on a large real road network and extensive experiments are tested for analysis. Qian and Eglese (2016) proposed a column generation based tabu search method to solve the time-dependent emission-minimizing routing problem, which implicitly considered the path selection in the pricing subproblems. Besides, Qian and Eglese (2016) use the NAEI emission model, which is independent of the vehicle load.

3. Problem description

For the convenience of the reader, we provide Table 1 listing the notation used in this paper.

A logistics company operates a homogeneous fleet of vehicles to deliver goods to customers that are geographically scattered in an urban area. The deliveries are subject to vehicle capacity and delivery time horizon restrictions. The objective is to minimize the operational cost, including fuel consumption cost and vehicle depreciation cost. In the logistics practice in China, a logistics company usually pays drivers with a fixed base salary plus the bonus based on the delivery quantity. Drivers' cost is therefore independent of the routing decision and is a constant. Thus, we do not consider the drivers' cost in the objective function.

Table 1
Notation.

Notation	Description	First appeared section
G^c	Customer graph	3
N^c	Set of customer nodes	3
A^c	Set of arcs connecting pairs of customer nodes	3
G^g	Geographical graph	3
N^g	Set of intersections	3
A^g	Set of road segments	3
$[L, U]$	Delivery time horizon	3
P_{ij}	Set of paths connecting arc $(i, j) \in A^c$	3
$\mathcal{P}_{ij}^p$	A path in the set P_{ij}	3
q_i	Demand of customer node $i \in N^c \setminus \{0\}$	3
s_i	Service time of customer node $i \in N^c \setminus \{0\}$	3
K	Number of vehicles	3
Q	Capacity of a vehicle	3
R	Number of time periods with constant speed	3
ψ_e, ψ_s, ψ_w	Coefficients in CMEM model to calculate fuel consumption	4.1
μ	Vehicle curb-weight	4.1
$F(t, v, f)$	Fuel consumption function of constant speed v , duration t , and vehicle load f	4.1
v_{ij}^{pr}	Speed in time period r on path $\mathcal{P}_{ij}^p$	4.2
w	Departure time from a node	4.2
$\tau_{ij}^p(w)$	Piecewise linear travel time function of departure time w from node i on path $\mathcal{P}_{ij}^p$	4.2
M_{ij}^p	Intervals of the piecewise linear travel time function $\tau_{ij}^p(w)$ on path $\mathcal{P}_{ij}^p$	4.2
$[b_{ij}^{pm}, b_{ij}^{p,m+1}]$	m th interval of travel time function $\tau_{ij}^p(w)$ on path $\mathcal{P}_{ij}^p$	4.2
t_{ij}^{pmr}	Travel time at speed v_{ij}^{pr} if the vehicles departs from node i at time b_{ij}^{pm} on path $\mathcal{P}_{ij}^p$	4.2
$\tau_{ij}^{pr}(w)$	Piecewise linear travel time function of departure time w at speed v_{ij}^{pr} on path $\mathcal{P}_{ij}^p$	4.2
θ_{ij}^{pmr}	Slope of piecewise linear function $\tau_{ij}^{pr}(w)$ within interval $[b_{ij}^{pm}, b_{ij}^{p,m+1}]$ on path $\mathcal{P}_{ij}^p$	4.2
η_{ij}^{pmr}	Intercept of piecewise linear function $\tau_{ij}^{pr}(w)$ within interval $[b_{ij}^{pm}, b_{ij}^{p,m+1}]$ on path $\mathcal{P}_{ij}^p$	4.2
d_{ij}^p	Distance of path $\mathcal{P}_{ij}^p$	4.2
f_{ij}^p	Vehicle load carried on path $\mathcal{P}_{ij}^p$	4.2
$F_{ij}^p(w)$	Fuel consumption of departure time w from node i on path $\mathcal{P}_{ij}^p$	4.2
S	Number of discretized time points to calculate the time-dependent shortest path	4.3
x_{ij}	Equal to 1 if arc (i, j) is on the optimal route	5.1
x_{ij}^p	Equal to 1 if the vehicle travels on path $\mathcal{P}_{ij}^p$	5.1
x_{ij}^{pm}	Equal to 1 if the departure time at node i toward node j is in $[b_{ij}^{pm}, b_{ij}^{p,m+1}]$	5.1
f_{ij}	Load carried on arc (i, j)	5.1
ϑ_{ij}^{pm}	Earliest possible departure time at node i on path $\mathcal{P}_{ij}^p$	5.1
ζ_{ij}^{pm}	Time flexibility at node i on path $\mathcal{P}_{ij}^p$	5.1
w_{ij}^{pm}	Actual departure time at node i (in $[b_{ij}^{pm}, b_{ij}^{p,m+1}]$) on path $\mathcal{P}_{ij}^p$	5.1
c^f	Unit cost of fuel consumption	5.1
c^d	Unit cost of vehicle depreciation	5.1

The traffic condition of the urban road network varies both temporally and spatially. Therefore, when travelling between two locations in the city at different time of the day, a vehicle may choose a different path, depending on the traffic condition.

The customer locations and the road network can be represented by two layers of a graph: customer graph and geographical graph (Fig. 3). Following traditional settings of vehicle routing problems, in the *customer graph* $G^c = (N^c, A^c)$, the node set N^c represents the set of customer locations and the depot location, while the directed arc set A^c connects the node pairs. Specifically, node $0 \in N^c$ represents the depot. Each customer node $i \in N^c \setminus \{0\}$ is associated with a demand q_i and service time s_i . The homogenous fleet has K vehicles each with a capacity Q . Vehicles are routed to satisfy all the customer demands within the delivery time horizon $[L, U]$.

The *geographical graph* $G^g = (N^g, A^g)$ represents the actual road network underlying the customer graph G^c . N^g denotes the set of intersections and A^g denotes the set of road segments in the road network. Each arc $(i, j) \in A^c$ in the customer graph can correspond to a set of multiple paths in the geographical graph, denoted as $P_{ij} = \{\mathcal{P}_{ij}^1, \mathcal{P}_{ij}^2, \dots\}$. A path $\mathcal{P}_{ij}^p$, $p = 1, \dots, |P_{ij}|$, connects nodes i and j in N^c , and $\mathcal{P}_{ij}^p$ consists of a finite sequence of road segments in A^g , denoted as $\mathcal{P}_{ij}^p = \{a_1^p, a_2^p, \dots\}$.

Fig. 4 shows the speed patterns of the road network in Beijing from October 14th to 18th (Monday to Friday), 2013. While the speed may vary in region or district, the speed patterns of the entire road network remain similar. Therefore, we partition the time of the day into R periods, and assume that, within each time period $r = 1, \dots, R$, the average speed of a road segment a remains stable, denoted as v_a^r .

Besides path flexibility, we also study the effect of *time flexibility*, which refers to the flexible departure time after L at the depot ($i = 0$) and/or the *post-service* waiting time at a customer node $i \in N^c \setminus \{0\}$, to avoid congestion on the road. The time flexibility is bounded. For example, the post-service waiting time at a customer node is normally subject to the time

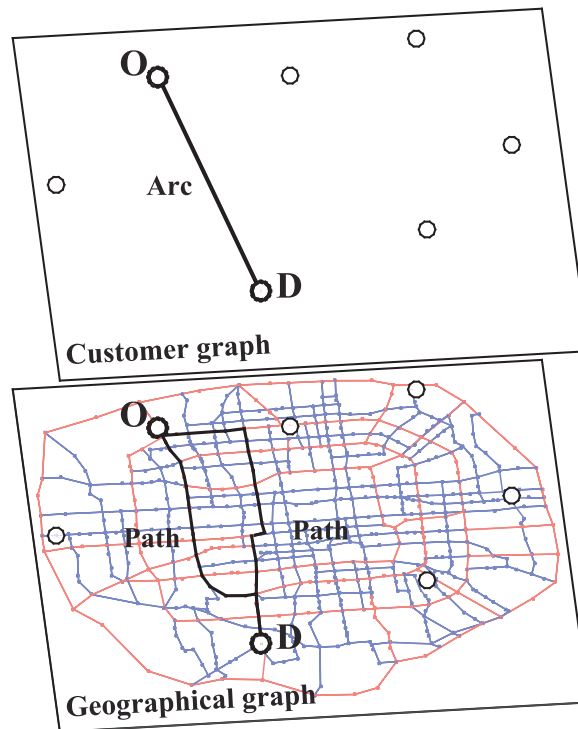


Fig. 3. Customer graph and geographical graph.

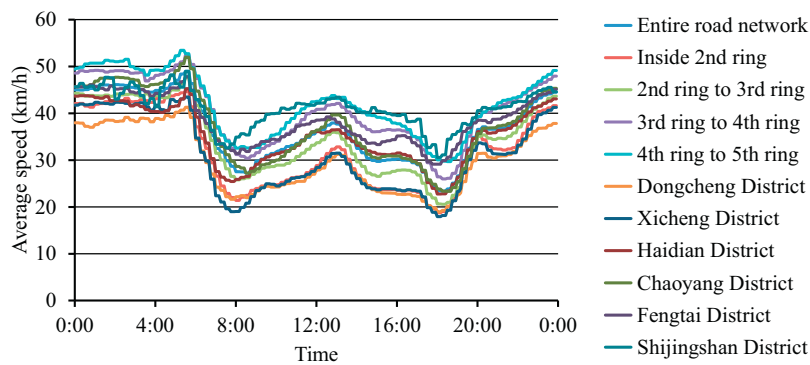


Fig. 4. Speed patterns of the road network in Beijing (Source: Beijing Municipal Commission of Transport).

restriction of the loading/unloading zone. Note also that time flexibility is different from the waiting time before the start of service at a customer node in the case of early arrival in the vehicle routing problem with time windows (VRPTW).

Furthermore, the traffic condition of the road network is also stochastic. That is, when the logistics company makes the routing decisions at the beginning of the time horizon, the actual road condition (or congestion level) of a particular road segment at a particular time period remains uncertain.

4. Preliminaries

In this section, we describe the fundamentals of the fuel consumption model, the structure of the time-dependent travel time model, and a method to obtain the time-dependent shortest paths. This section basically prepares for our model formulation in Section 5.

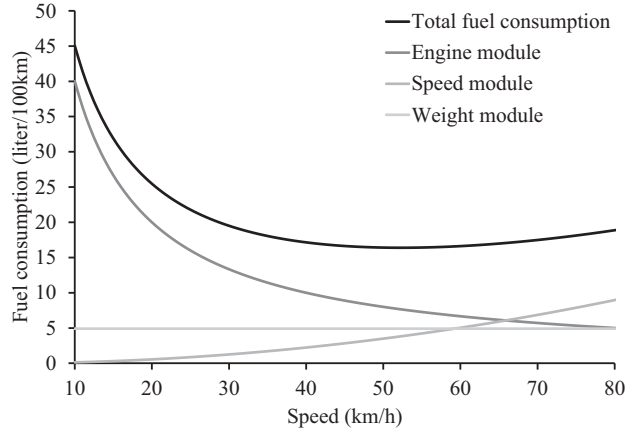


Fig. 5. Fuel consumption function of speed (with 2000 kg load).

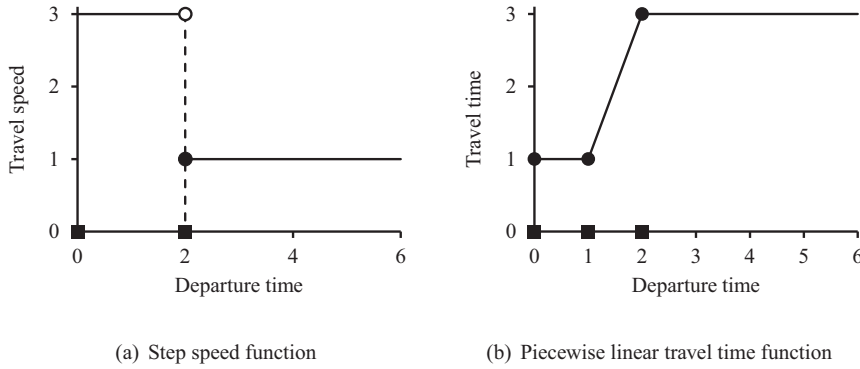


Fig. 6. An example of converting step speed function to piecewise linear travel time function on a road segment, with 3 units of distance.

4.1. Fuel consumption calculation

Same as relevant papers in the literature (e.g., Bektaş and Laporte, 2011; Franceschetti et al., 2013), we also use the Comprehensive Modal Emission Model (CMEM, Barth et al., 2005 and Barth and Boriboonsomsin, 2009) to calculate fuel consumption.

For a vehicle traveling a distance d at a constant speed v in duration t ($t = d/v$) and carrying a load f , the fuel consumption $F(t, v, f)$ can be calculated as

$$F(t, v, f) = \psi_e t + \psi_s v^3 t + \psi_w d(\mu + f), \quad (1)$$

where μ is the vehicle curb-weight, and ψ_e , ψ_s , and ψ_w are the coefficients of the engine, speed, and weight modules, respectively. Fig. 5 illustrates the fuel consumption function of a typical light-duty vehicle in China (parameters and values are in Table A.16). The function has a U-shape with the minimum fuel consumption at the speed of 52.27 km/h.

Detailed description of the CMEM is in Appendix A.

4.2. Time-dependent travel time model

As discussed in Section 3, we assume that the delivery time horizon $[L, U]$ consists of R time periods and the travel speed of a particular road segment a is constant within a period r . Thus the travel speed of road segment a is a step function of the time of the day. The corresponding travel time function of road segment a is a piecewise linear function, where we use the same method as described in Ichoua et al. (2003) for the conversion from the step speed function to the piecewise linear travel time function on a road segment, illustrated in Fig. 6. For a path $\mathcal{P}_{ij}^p$ with a finite sequence of road segments, each of which has a continuous piecewise linear travel time function, we use the method described in Ghiani and Guerriero (2014) to generate $\tau_{ij}^p(\cdot)$, the travel time function on the entire path $\mathcal{P}_{ij}^p$, which is also a continuous piecewise linear function.

Considering the piecewise linear travel time function $\tau_{ij}^p(\cdot)$ of path $\mathcal{P}_{ij}^p$ of arc (i, j) , it consists of M_{ij}^p intervals defined by a finite set of break points, $[b_{ij}^{pm}, b_{ij}^{p,m+1}]$, $m = 1, \dots, M_{ij}^p$. Within each interval the travel time function $\tau_{ij}^p(\cdot)$ is linear. Note

that $b_{ij}^{p1} = L$ and $b_{ij}^{p, M_{ij}^p+1} = U$, as the domain (set of inputs) of the travel time function $\tau_{ij}^p(\cdot)$ is the delivery time horizon $[L, U]$.

Let t_{ij}^{pmr} denote the travel time the vehicle travels at speed v_{ij}^{pr} along the path $\mathcal{P}_{ij}^p$, if it departs from node i at time b_{ij}^{pm} . If the vehicle departs from node i at time $w \in [b_{ij}^{pm}, b_{ij}^{p, m+1}]$, the travel time it travels at speed v_{ij}^{pr} is

$$\begin{aligned}\tau_{ij}^{pr}(w) &= \frac{t_{ij}^{p, m+1, r} - t_{ij}^{pmr}}{b_{ij}^{p, m+1} - b_{ij}^{pm}} (w - b_{ij}^{pm}) + t_{ij}^{pmr} \\ &= \theta_{ij}^{pmr} w + \eta_{ij}^{pmr},\end{aligned}\quad (2)$$

where

$$\theta_{ij}^{pmr} = \frac{t_{ij}^{p, m+1, r} - t_{ij}^{pmr}}{b_{ij}^{p, m+1} - b_{ij}^{pm}}, \quad \text{and} \quad (3)$$

$$\eta_{ij}^{pmr} = t_{ij}^{pmr} - b_{ij}^{pm} \frac{t_{ij}^{p, m+1, r} - t_{ij}^{pmr}}{b_{ij}^{p, m+1} - b_{ij}^{pm}}. \quad (4)$$

Clearly, $t_{ij}^{pmr} = \tau_{ij}^{pr}(b_{ij}^{pm})$. The total travel time on path $\mathcal{P}_{ij}^p$ is then

$$\tau_{ij}^p(w) = \sum_{r=1}^R \tau_{ij}^{pr}(w) = \sum_{r=1}^R \theta_{ij}^{pmr} w + \sum_{r=1}^R \eta_{ij}^{pmr}. \quad (5)$$

Therefore, the fuel consumption of a vehicle traveling on path $\mathcal{P}_{ij}^p$ of arc (i, j) and departing from node i at time $w \in [b_{ij}^{pm}, b_{ij}^{p, m+1}]$ can be calculated as

$$F_{ij}^p(w) = \psi_e \left[\sum_{r \in R} \theta_{ij}^{pmr} w + \sum_{r \in R} \eta_{ij}^{pmr} \right] + \psi_s \left[\sum_{r \in R} (v_{ij}^{pr})^3 \theta_{ij}^{pmr} w + \sum_{r \in R} (v_{ij}^{pr})^3 \eta_{ij}^{pmr} \right] + \psi_w d_{ij}^p (\mu + f_{ij}^p), \quad (6)$$

where d_{ij}^p denotes the distance of path $\mathcal{P}_{ij}^p$ and f_{ij}^p denotes the load carried by the vehicle on path $\mathcal{P}_{ij}^p$ of arc (i, j) .

4.3. Time-dependent shortest path method

In a time-dependent network where the travel time function of each arc is piecewise linear, the goal of the *time-dependent shortest path problem* is to find the shortest paths from i to j (measured by travel time), for all possible departure time at node i . This problem is NP-hard in its general form (Dehne et al., 2012). However, for a particular given departure time w at node i , we can use a modified Dijkstra's algorithm (Algorithm 1) to find the time-dependent shortest path from node i to j efficiently (Brodal and Jacob, 2004). Note that, in Step 2.2(a), we calculate the value T_n based on the time-dependent travel time function $\tau_{n'n}(V_{n'})$.

Algorithm 1 Modified Dijkstra's algorithm for the time-dependent shortest path on arc (i, j) .

Step 1. Initialization.

- 1.1. Label the origin customer node i as $[0, -]$, where $[V_i, n_i^{pre}]$ represents the value V_i and the predecessor node n_i^{pre} of node i .
- 1.2. Label all other nodes in N^g as $[\infty, -]$.
- 1.3. Set customer node i as "scanned."

Step 2. Label update.

- 2.1. Denote the last scanned node n' .
- 2.2. $\forall (n', n) \in A^g$ and $n \in N^g$ not scanned,
 - a) Compute $T_n = V_{n'} + \tau_{n'n}(V_{n'})$, where $\tau_{n'n}(V_{n'})$ represents the travel time on the road segment (n', n) when the vehicle departs from node n' at time $V_{n'}$.
 - b) If $T_n < V_n$, relabel node n with $[T_n, n']$.

Step 3. Scan a node.

Find the unscanned node with the smallest V_n and scan the node.

Step 4. Termination check.

If the customer node j is scanned, stop.

Else, go to step 2.

In this paper, we solve the time-dependent shortest path problem by [Algorithm 1](#) with S discretized time points $w = w_1, \dots, w_S$ as the given departure time at node i , to collect a set of distinctive time-dependent shortest paths (time-minimizing paths), $P_{ij} = \{\mathcal{P}_{ij}^1, \mathcal{P}_{ij}^2, \dots\}$. Further, besides the time-minimizing paths, we also include the distance-minimizing path on arc (i, j) in P_{ij} . The set of paths serve as candidate paths to be selected when the vehicle travels on arc (i, j) .

5. Mathematical models

In this section, we formulate the deterministic time-dependent vehicle routing problem with path flexibility (TDVRP-PF) as a mixed integer program and the stochastic TDVRP-PF as a two-stage stochastic mixed integer program.

5.1. Deterministic TDVRP-PF

Under deterministic traffic conditions, the vehicle routing and path selection decisions are made simultaneously to minimize the operational cost, that is, the fuel consumption and vehicle depreciation cost. The set of candidate paths P_{ij} of each arc $(i, j) \in A^c$ are preprocessed as discussed in [Section 4.3](#).

We define the decision variables as

$$x_{ij} = \begin{cases} 1, & \text{if arc } (i, j) \text{ is on the optimal route,} \\ 0, & \text{otherwise.} \end{cases}$$

$$x_{ij}^p = \begin{cases} 1, & \text{if the vehicle travels on path } \mathcal{P}_{ij}^p, \\ 0, & \text{otherwise.} \end{cases}$$

$$x_{ij}^{pm} = \begin{cases} 1, & \text{if the departure time at node } i \text{ toward node } j \text{ is in } [b_{ij}^{pm}, b_{ij}^{p,m+1}], \\ 0, & \text{otherwise.} \end{cases}$$

$$f_{ij} = \text{load carried on arc } (i, j).$$

$$f_{ij}^p = \text{load carried on path } \mathcal{P}_{ij}^p.$$

$$\vartheta_{ij}^{pm} = \text{earliest possible departure time at node } i \text{ on path } \mathcal{P}_{ij}^p.$$

$$\zeta_{ij}^{pm} = \text{time flexibility at node } i \text{ on path } \mathcal{P}_{ij}^p.$$

$$w_{ij}^{pm} = \vartheta_{ij}^{pm} + \zeta_{ij}^{pm} = \text{actual departure time at node } i \text{ (in } [b_{ij}^{pm}, b_{ij}^{p,m+1}]) \text{ on path } \mathcal{P}_{ij}^p.$$

We introduce x_{ij}^p and f_{ij}^p to specify the path selection on arc (i, j) , and introduce x_{ij}^{pm} , w_{ij}^{pm} , ϑ_{ij}^{pm} , and ζ_{ij}^{pm} to specify the departure time at node i within a time interval $[b_{ij}^{pm}, b_{ij}^{p,m+1}]$ of the piecewise linear travel time function along path $\mathcal{P}_{ij}^p$.

Note that when we preprocess the set of candidate paths P_{ij} for each arc $(i, j) \in A^c$, the information related to the underlying road network $G^s = (N^s, A^s)$ is all embedded in P_{ij} . Therefore, in the TDVRP-PF, we focus on the customer graph $G^c = (N^c, A^c)$ and the explicit candidate set of paths P_{ij} on each arc $(i, j) \in A^c$, obtained from the geographical graph. For model simplicity, we drop the superscript “c” and use $G = (N, A)$ to represent the customer graph $G^c = (N^c, A^c)$. Further, for each arc $(i, j) \in A$, there are $|P_{ij}|$ candidate paths, and for each path $\mathcal{P}_{ij}^p \in P_{ij}$, there are $|M_{ij}^p|$ intervals to represent the piecewise linear travel time function. In the following, in order to simplify the notation, we drop the super/subscripts and use P for P_{ij} and M for M_{ij}^p .

The TDVRP-PF is formulated as a mixed integer program as follows.

$$\text{Min} \quad (TDVRP-PF)$$

$$c^f \left\{ \sum_{(i,j) \in A} \sum_{p \in P} \sum_{m \in M} \psi_e \left(\sum_{r \in R} \theta_{ij}^{pmr} w_{ij}^{pm} + \sum_{r \in R} \eta_{ij}^{pmr} x_{ij}^{pm} \right) \right. \quad (7)$$

$$+ \sum_{(i,j) \in A} \sum_{p \in P} \sum_{m \in M} \psi_s \left[\sum_{r \in R} (v_{ij}^{pr})^3 \theta_{ij}^{pmr} w_{ij}^{pm} + \sum_{r \in R} (v_{ij}^{pr})^3 \eta_{ij}^{pmr} x_{ij}^{pm} \right] \quad (8)$$

$$+ \sum_{(i,j) \in A} \sum_{p \in P} \psi_w d_{ij}^p (\mu x_{ij}^p + f_{ij}^p) \left. \right\} \quad (9)$$

$$+ c^d \sum_{(i,j) \in A} \sum_{p \in P} d_{ij}^p x_{ij}^p \quad (10)$$

s.t.

$$\sum_{j \in N} x_{0j} \leq K, \quad (11)$$

$$\sum_{i \in N} x_{ij} = 1, \quad \forall j \in N \setminus \{0\}, \quad (12)$$

$$\sum_{j \in N} x_{ij} = 1, \quad \forall i \in N \setminus \{0\}, \quad (13)$$

$$\sum_{i \in N} f_{ij} - \sum_{k \in N} f_{jk} = q_j, \quad \forall j \in N \setminus \{0\}, \quad (14)$$

$$q_j x_{ij} \leq f_{ij} \leq (Q - q_i) x_{ij}, \quad \forall (i, j) \in A, \quad (15)$$

$$\sum_{p \in P} x_{ij}^p = x_{ij}, \quad \forall (i, j) \in A, \quad (16)$$

$$\sum_{m \in M} x_{ij}^{pm} = x_{ij}^p, \quad \forall (i, j) \in A, \forall p \in P, \quad (17)$$

$$\sum_{p \in P} f_{ij}^p = f_{ij}, \quad \forall (i, j) \in A, \quad (18)$$

$$q_j x_{ij}^p \leq f_{ij}^p \leq (Q - q_i) x_{ij}^p, \quad \forall (i, j) \in A, \forall p \in P, \quad (19)$$

$$\vartheta_{0j}^{pm} = L x_{0j}^{pm}, \quad \forall j \in N \setminus \{0\}, \forall p \in P, \forall m \in M, \quad (20)$$

$$\vartheta_{ij}^{pm} + \zeta_{ij}^{pm} = w_{ij}^{pm}, \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M, \quad (21)$$

$$\zeta_{ij}^{pm} \leq u_i, \quad \forall i \in N, \forall p \in P, \forall m \in M, \quad (22)$$

$$b_{ij}^{pm} x_{ij}^{pm} \leq w_{ij}^{pm} \leq b_{ij}^{p,m+1} x_{ij}^{pm}, \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M, \quad (23)$$

$$\begin{aligned} & \sum_{i \in N} \sum_{p \in P} \sum_{m \in M} \left(\sum_{r \in R} \theta_{ij}^{pmr} w_{ij}^{pm} + \sum_{r \in R} \eta_{ij}^{pmr} x_{ij}^{pm} \right) \\ & + \sum_{i \in N} \sum_{p \in P} \sum_{m \in M} w_{ij}^{pm} + s_j = \sum_{k \in N} \sum_{p \in P} \sum_{m \in M} \vartheta_{jk}^{pm}, \quad \forall j \in N \setminus \{0\}, \end{aligned} \quad (24)$$

$$\sum_{p \in P} \sum_{m \in M} \left(\sum_{r \in R} \theta_{i0}^{pmr} w_{i0}^{pm} + \sum_{r \in R} \eta_{i0}^{pmr} x_{i0}^{pm} \right) \leq U, \quad \forall i \in N \setminus \{0\}, \quad (25)$$

$$x_{ij} \in \{0, 1\}, \quad \forall (i, j) \in A, \quad (26)$$

$$x_{ij}^p \in \{0, 1\}, \quad \forall (i, j) \in A, \forall p \in P, \quad (27)$$

$$x_{ij}^{pm} \in \{0, 1\}, \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M, \quad (28)$$

$$f_{ij} \geq 0, \quad \forall (i, j) \in A, \quad (29)$$

$$f_{ij}^p \geq 0, \quad \forall (i, j) \in A, \forall p \in P, \quad (30)$$

$$w_{ij}^{pm} \geq 0, \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M, \quad (31)$$

$$\vartheta_{ij}^{pm}, \zeta_{ij}^{pm} \geq 0, \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M. \quad (32)$$

In the objective function, expressions (7), (8), and (9) are the engine, speed, and weight modules of the fuel cost, and expression (10) is the distance-based vehicle depreciation cost. c^f and c^d are the corresponding cost coefficients.

Constraints (11)–(15) are standard constraints of the two-index single-commodity flow model of the capacitated vehicle routing problem (Gavish and Graves, 1981). Specifically, constraint (11) imposes that there are no more than K vehicles, constraints (12) and (13) are the vehicle flow conservation constraints, constraints (14) are the commodity flow conservation constraints, and constraints (15) guarantee the capacity of each vehicle is not exceeded.

Constraints (16) restrict the vehicle to select exactly one path, if traveling on arc $(i, j) \in A$, and constraints (17) restrict the vehicle to depart at node i within exactly one time interval on the selected path $\mathcal{P}_{ij}^p$. Constraints (18) and (19) ensure that the commodity is carried on exactly one path and vehicle capacity is not exceeded, if the vehicle travels on arc $(i, j) \in A$.

Constraints (20) specify that the vehicle departs from the depot no earlier than the beginning of the time horizon. Constraints (21) calculate the actual departure time at the depot ($i = 0$) or a customer node $i \in N^c \setminus \{0\}$, if time flexibility (i.e., flexible departure time at the depot and/or post-service waiting time at customer sites) is allowed. Constraints (22) ensure the flexible departure time at the depot ($i = 0$) and the post-service waiting time at customer site $i \in N^c \setminus \{0\}$ do not exceed their corresponding upper bounds u_i . Constraints (23) restrict the departure time at node i on the selected path $\mathcal{P}_{ij}^p$ to be within the time interval as specified in (17). If path $\mathcal{P}_{ij}^p$ is not selected, $w_{ij}^{pm} = 0, \forall m \in M$. Constraints (24) require that, if the vehicle travels on path $\mathcal{P}_{ij}^p$ of arc (i, j) , the earliest possible departure time at customer node j is the summation of the actual departure time at customer node i , the travel time on path $\mathcal{P}_{ij}^p$, and the service time at customer node j , which ensures the time continuity of a vehicle traveling in the network. Constraints (25) require that the delivery is completed within the delivery time horizon.

Last, constraints (26)–(32) are variable specification constraints. Note that constraints (15) become redundant due to constraints (18) and (19), but we keep them here for clarity.

When time flexibility is not allowed, $u_i = 0$ in constraints (22), we have $\zeta_{ij}^{pm} = 0$ and constraints (21) reduce to $\vartheta_{ij}^{pm} = w_{ij}^{pm}$. We can replace ϑ_{ij}^{pm} with w_{ij}^{pm} in constraints (20) and (24), eliminate decision variables ϑ_{ij}^{pm} , and eliminate constraints (21), (22), and (32).

When the set of candidate paths P_{ij} of each arc $(i, j) \in A^c$ reduces to one path, the model TDVRP–PF reduces to the time-dependent vehicle routing problem *without* path flexibility (TDVRP). We drop the superscript p in the TDVRP–PF model. That is, $x_{ij}^p, f_{ij}^p, x_{ij}^{pm}, w_{ij}^{pm}, \vartheta_{ij}^{pm}$, and ζ_{ij}^{pm} become $x_{ij}, f_{ij}, x_{ij}^m, w_{ij}^m, \vartheta_{ij}^m$, and ζ_{ij}^m , respectively. Consequently, we can eliminate constraints (16), (18), (19), (27), and (30), and then drop the superscript p in the objective function and the remaining constraints.

5.2. Stochastic TDVRP–PF

To deal with stochastic traffic conditions in the road network, we formulate a two-stage stochastic mixed integer program, combining the vehicle routing decisions based on statistical analysis of the expected traffic patterns and the en route path selection decisions based on real-time traffic conditions.

We model the stochastic traffic conditions as a finite set of scenarios Ω . Denote $\omega \in \Omega$ as a random scenario and denote $\xi(\omega) = (\theta_{ij}^{pmr}(\omega), \eta_{ij}^{pmr}(\omega), v_{ij}^{pr}(\omega), b_{ij}^{pm}(\omega))$ as the vector of the random variables representing the traffic condition under scenario ω . In the two-stage stochastic program, the vehicle routing decisions $\mathbf{x} = (x_{ij}, f_{ij})$ are the first-stage decisions. Given the scenario $\omega \in \Omega$, the path selection and related traveling decisions $\mathbf{y}(\omega) = (x_{ij}^p(\omega), x_{ij}^{pm}(\omega), f_{ij}^p(\omega), w_{ij}^{pm}(\omega), \vartheta_{ij}^{pm}(\omega), \zeta_{ij}^{pm}(\omega))$ are the second-stage decisions. The recourse decisions thus involve the path selection.

The two-stage stochastic mixed integer program (TDVRP-PF^S) is formulated as follows:

$$\min E_{\xi}[h(\mathbf{x}, \xi)] \quad (33)$$

s.t.

$$\sum_{j \in N} x_{0j} \leq K, \quad (34)$$

$$\sum_{i \in N} x_{ij} = 1, \quad \forall j \in N \setminus \{0\}, \quad (35)$$

$$\sum_{j \in N} x_{ij} = 1, \quad \forall i \in N \setminus \{0\}, \quad (36)$$

$$\sum_{i \in N} f_{ij} - \sum_{k \in N} f_{jk} = q_j, \quad \forall j \in N \setminus \{0\}, \quad (37)$$

$$q_j x_{ij} \leq f_{ij} \leq (Q - q_i) x_{ij}, \quad \forall (i, j) \in A, \quad (38)$$

$$x_{ij} \in \{0, 1\}, \quad \forall (i, j) \in A, \quad (39)$$

$$f_{ij} \geq 0, \quad \forall (i, j) \in A, \quad (40)$$

where for each $\omega \in \Omega$,

$$h(\mathbf{x}, \xi(\omega)) = \min_{\mathbf{y}(\omega)} c^f \left\{ \sum_{(i,j) \in A} \sum_{p \in P} \sum_{m \in M} \psi_e \left[\sum_{r \in R} \theta_{ij}^{pmr}(\omega) w_{ij}^{pm}(\omega) + \sum_{r \in R} \eta_{ij}^{pmr}(\omega) x_{ij}^{pm}(\omega) \right] \right. \quad (41)$$

$$+ \sum_{(i,j) \in A} \sum_{p \in P} \sum_{m \in M} \psi_s \left[\sum_{r \in R} (v_{ij}^{pr}(\omega))^3 \theta_{ij}^{pmr}(\omega) w_{ij}^{pm}(\omega) + \sum_{r \in R} (v_{ij}^{pr}(\omega))^3 \eta_{ij}^{pmr}(\omega) x_{ij}^{pm}(\omega) \right] \quad (42)$$

$$\left. + \sum_{(i,j) \in A} \sum_{p \in P} \psi_w d_{ij}^p (\mu x_{ij}^p(\omega) + f_{ij}^p(\omega)) \right\} \quad (43)$$

$$+ c^d \sum_{(i,j) \in A} \sum_{p \in P} d_{ij}^p x_{ij}^p(\omega) \quad (44)$$

s.t.

$$\sum_{p \in P} x_{ij}^p(\omega) = x_{ij}, \quad \forall (i, j) \in A, \quad (45)$$

$$\sum_{m \in M} x_{ij}^{pm}(\omega) = x_{ij}^p(\omega), \quad \forall (i, j) \in A, \forall p \in P, \quad (46)$$

$$\sum_{p \in P} f_{ij}^p(\omega) = f_{ij}, \quad \forall (i, j) \in A, \quad (47)$$

$$q_j x_{ij}^p(\omega) \leq f_{ij}^p(\omega) \leq (Q - q_i) x_{ij}^p(\omega), \quad \forall (i, j) \in A, \forall p \in P, \quad (48)$$

$$\vartheta_{0j}^{pm}(\omega) = L x_{0j}^{pm}(\omega), \quad \forall j \in N \setminus \{0\}, \forall p \in P, \forall m \in M, \quad (49)$$

$$\vartheta_{ij}^{pm}(\omega) + \zeta_{ij}^{pm}(\omega) = w_{ij}^{pm}(\omega), \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M, \quad (50)$$

$$\zeta_{ij}^{pm}(\omega) \leq u_i, \quad \forall i \in N, \forall p \in P, \forall m \in M, \quad (51)$$

$$b_{ij}^{pm}(\omega)x_{ij}^{pm}(\omega) \leq w_{ij}^{pm}(\omega) \leq b_{ij}^{p,m+1}(\omega)x_{ij}^{pm}(\omega), \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M, \quad (52)$$

$$\begin{aligned} & \sum_{i \in N} \sum_{p \in P} \sum_{m \in M} \left[\sum_{r \in R} \theta_{ij}^{pmr}(\omega) w_{ij}^{pm}(\omega) + \sum_{r \in R} \eta_{ij}^{pmr}(\omega) x_{ij}^{pm}(\omega) \right] \\ & + \sum_{i \in N} \sum_{p \in P} \sum_{m \in M} w_{ij}^{pm}(\omega) + s_j = \sum_{k \in N} \sum_{p \in P} \sum_{m \in M} \vartheta_{jk}^{pm}(\omega), \quad \forall j \in N \setminus \{0\}, \end{aligned} \quad (53)$$

$$\sum_{p \in P} \sum_{m \in M} \left[\sum_{r \in R} \theta_{i0}^{pmr}(\omega) w_{i0}^{pm}(\omega) + \sum_{r \in R} \eta_{i0}^{pmr}(\omega) x_{i0}^{pm}(\omega) \right] \leq U, \quad \forall i \in N \setminus \{0\}, \quad (54)$$

$$x_{ij}^p(\omega) \in \{0, 1\}, \quad \forall (i, j) \in A, \forall p \in P, \quad (55)$$

$$x_{ij}^{pm}(\omega) \in \{0, 1\}, \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M, \quad (56)$$

$$f_{ij}^p(\omega) \geq 0, \quad \forall (i, j) \in A, \forall p \in P, \quad (57)$$

$$w_{ij}^{pm}(\omega) \geq 0, \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M, \quad (58)$$

$$\vartheta_{ij}^{pm}(\omega), \zeta_{ij}^{pm}(\omega) \geq 0, \quad \forall (i, j) \in A, \forall p \in P, \forall m \in M. \quad (59)$$

The constraints are similar to the deterministic model (7)–(32), except that the scenarios are introduced to the second-stage subproblems. The first-stage master problem (33)–(40) is a capacitated vehicle routing problem with the objective function considering the expected cost of the second-stage subproblems (41)–(59). The stochastic program (TDVRP-PF^S) has mixed-integer variables in both stages, and the recourse matrix is random, which makes it very difficult to solve.

When time flexibility (i.e., flexible departure time at the depot and/or post-service waiting time at customer sites) is not allowed, the simplification of the TDVRP-PF^S is similar to that in the deterministic model. When the set of candidate paths P_{ij} of each arc $(i, j) \in A^c$ reduces to one path, the model TDVRP-PF^S reduces to the stochastic time-dependent vehicle routing problem *without* path flexibility (TDVRP^S), where we drop the superscript p in the second-stage subproblems, similar to the conversion from the TDVRP-PF to the TDVRP. However, there is no actual recourse action in the second-stage subproblems, due to the lack of path flexibility. Therefore, the second-stage subproblems solely serve as the evaluation of the first-stage routing decisions under stochastic traffic conditions.

In Table 2, we give the problem sizes of both TDVRP-PF and TDVRP-PF^S. To be specific, in the deterministic TDVRP-PF, the integer variables consist of x_{ij} , x_{ij}^p , and x_{ij}^{pm} , while the continuous variables consist of f_{ij} , f_{ij}^p , w_{ij}^{pm} , ϑ_{ij}^{pm} , and ζ_{ij}^{pm} . In the TDVRP-PF^S, the integer and continuous variables are similar, except that the scenarios are introduced in the second-stage variables, i.e., $\mathbf{y}(\omega) = (x_{ij}^p(\omega), x_{ij}^{pm}(\omega), f_{ij}^p(\omega), w_{ij}^{pm}(\omega), \vartheta_{ij}^{pm}(\omega), \zeta_{ij}^{pm}(\omega))$. Observe that the problem size of the subproblem of one scenario in the TDVRP-PF^S is similar to that of the deterministic TDVRP-PF, which is already large. Therefore, even if we limit the number of scenarios in the TDVRP-PF^S, the size of the stochastic program is still huge.

Table 2
Problem sizes of the TDVRP-PF and TDVRP-PF^S.

		# constraints	# integer var.	# continuous var.
Deterministic		$\mathcal{O}(N ^2 \cdot P \cdot M)$	$\mathcal{O}(N ^2 \cdot P \cdot M)$	$\mathcal{O}(N ^2 \cdot P \cdot M)$
Stochastic	Master problem	$\mathcal{O}(N ^2)$	$\mathcal{O}(N ^2)$	$\mathcal{O}(N ^2)$
	Subproblem (each scenario)	$\mathcal{O}(N ^2 \cdot P \cdot M)$	$\mathcal{O}(N ^2 \cdot P \cdot M)$	$\mathcal{O}(N ^2 \cdot P \cdot M)$

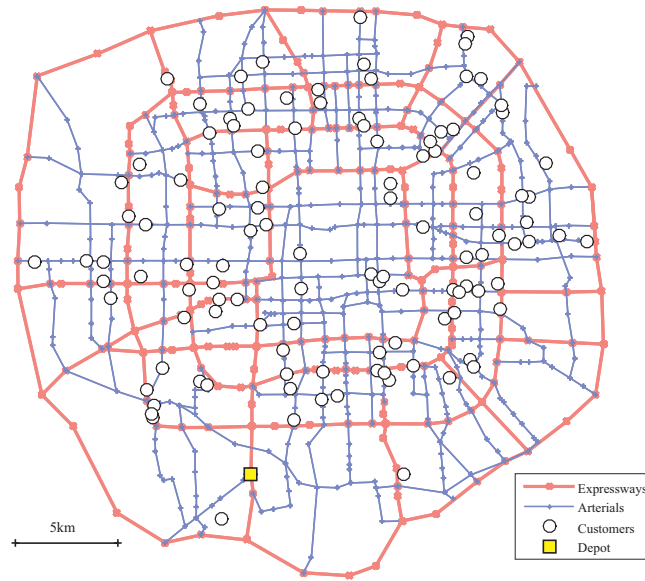


Fig. 7. Road network of the urban area in Beijing and candidate customer locations.

Table 3

Average speed of different road categories.

	Expressways	Arterials	Residential roads
06:00–07:00	48 km/h	32 km/h	24 km/h
07:00–09:00	36 km/h	24 km/h	24 km/h
09:00–13:00	41 km/h	28 km/h	24 km/h

6. Numerical experiment

In this section, we first describe the experimental settings in Section 6.1 and then provide the experimental results and analysis in Section 6.2.

6.1. Experimental design

Clearly, instance design is important. Similar to Qian and Eglese (2016) (for a London street network), we combined the traffic network from the Beijing situation and followed a similar approach as in Solomon (1987) for constructing the customer graph. In what follows, we describe the construction of the geographic graph, based on the road network of Beijing, and the generation of a testbed of instances. Further, we describe the experimental settings and the computational environment.

6.1.1. Network construction

We construct geographical graph (road network) based on the urban area of Beijing (within the 5th ring road). We classify the roads into three categories (Wang et al., 2008): *expressways*, including four ring roads (named from the 2nd to the 5th ring road in the order of radial distance from the city center) and urban freeways connecting the ring roads, *arterials*, including arterial and sub-arterial roads in the city, and *residential roads*, referring to relatively smaller roads, other than expressways and arterials.

In the urban area of Beijing, most roads are north-south or east-west. Therefore, we only depict expressways and arterials in the network and use Manhattan distances to calculate the traveling distances between locations along residential roads. The network of expressways and arterials in Beijing is shown in Fig. 7, consisting of 673 nodes and 1870 arcs.

The average speeds of different road categories are shown in Table 3. Note that, due to the spatial difference of congestion, speeds of the same road category vary in different areas of the city. The detailed speed setting is described in Appendix B. Note also that the traffic condition of residential roads is normally stable over the day and the speed is therefore assumed to be constant over time. People normally drive on these residential roads as the first mile and last mile connecting to expressways or arterials.

To construct the customer graph, we first generate a candidate customer set of 111 customer nodes (circles in Fig. 7), which are the locations of various retail stores such as supermarkets, fruit chains stores, and home appliance stores. In our

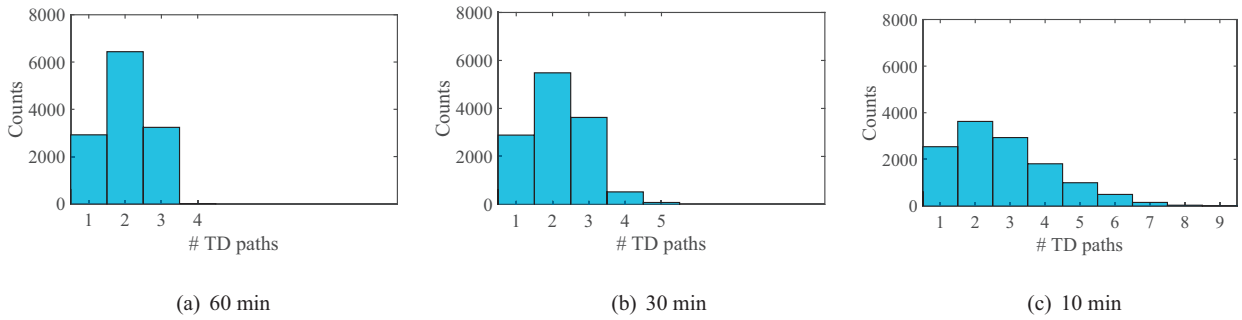


Fig. 8. Number of time-dependent paths under different discretization levels.

paper, they represent the customer locations of logistics companies in the city. The depot (the square node in Fig. 7) is located in the southern part of the city.

6.1.2. Test instances

We generate three types of instances, categorized similar to the R, C, and RC of the Solomon instances (Solomon, 1987). The R instances are the baseline instances. In each baseline instance, we randomly select 20 customers from the customer set. Each customer has an equal demand of 320 kg and the service time for each customer is 10 min. We generate 30 baseline instances, named R01, R02, ..., R30. The construction of the C and RC instances is described in Appendix C. We also provide examples of the R, C, and RC instances in Fig. C.12–C.14 in Appendix C.

Customers are served by a fleet of 2 homogeneous vehicles, each with a capacity of 4460 kg. Other parameters of the vehicles are listed in Table A.16 in Appendix A. The delivery time horizon (morning shift) is from 6:00 to 13:00. The cost of diesel c^f is 7.77 RMB/liter (approximately € 1.11/liter, or \$4.61/gallon) and the distance-based vehicle depreciation cost c^d is 0.45 RMB/km, (€ 0.064/km, or \$0.070/km). (The above calculation is based on the currency exchange rates, € 100 = 697.69 RMB and \$100 = 638.51 RMB, as of December 5, 2015, Bank of China.)

6.1.3. Experimental settings and computational environment

We first study the number of time-dependent paths in the TDVRP-PF as a trade-off between solution quality and computational effort. We then provide a detailed comparison between the TDVRP and TDVRP-PF, using the baseline instances. Further, we study the impact of time flexibility, customer network topology, and stochastic traffic conditions.

We analyze the deterministic TDVRP-PF based on 30 replications and the stochastic TDVRP-PF based on 10 replications. We solve the instances using IBM ILOG Cplex Optimization Studio 12.6 in the parallel mode up to 32 threads. Instances of the deterministic models are solved to optimality with the 0.1% optimality gap tolerance, while those of the stochastic models are solved to optimality with the 0.5% optimality gap tolerance. Except stated otherwise, we use the default values in Cplex. All experiments are conducted on a server with Intel E5 4640, 2.4 GHz CPUs and 64 × 8 GB memory.

6.2. Experimental results and analysis

We provide the experimental results and analysis. Except the experiments on customer network topology (Section 6.2.4), all experiments are based on the baseline (R1–R30) instances. Also, except the experiments on time flexibility (Section 6.2.3), all the experiments assume that vehicles leave the depot at 6:00 and post-service waiting time is not allowed at the customer sites.

6.2.1. Number of time-dependent paths

As described in Section 4.3, we use S discretized time points to generate candidate time-dependent shortest paths. The finer we discretize, the more candidate time-dependent shortest paths we may possibly generate, which offers more flexibility in the path selection, yet at the expense of more computational effort. We choose the discretized time points evenly in the delivery time horizon $[L, U]$ based on a *discretization level*, which is the time interval between two consecutive discretized time points. In the following, we compare the results of different discretization levels, i.e., 10 min, 30 min, and 60 min.

In Fig. 8, we show the arc count in the network of 30 baseline instances, based on the number of distinctive time-dependent shortest paths on each arc. For example, using the 60 min discretization level, there are six arcs with four distinctive time-dependent paths. There are 6438 arcs with two distinctive paths, accounting for 51.10% of the total number of arcs. (Note that there are $21 \times 20 = 420$ arcs in each instance and therefore $420 \times 30 = 12,600$ arcs in total in each histogram in Fig. 8.) We observe that the largest number of distinctive paths increases to five and nine, when the discretization level is set to 30 min and 10 min, respectively. However, the distributions have more arc counts on fewer paths and are skewed to the right (more paths) in all cases.

Table 4
TDVRP–PF under different discretization levels.

	Distance (km)	Fuel (liter)	Cost (RMB)	Soln. time (min)	# constraints	# integer var.	# continuous var.
60 min	127.94	23.83	243.20	348.34	38,481	12,217	22,763
30 min	127.94	23.83	243.20	407.88	40,614	12,901	24,083
10 min	127.80	23.81	243.02	593.37	48,613	15,544	29,146

Table 5
Comparison among CVRP, TDVRP, and TDVRP–PF.

Model	Distance		Fuel		Cost		Travel time	
	(km)	Diff. (%)	(liter)	Diff. (%)	(RMB)	Diff. (%)	(hour)	Diff. (%)
CVRP(+)	124.25	–	25.75	1.18%	256.52	0.93%	4.68	0.65%
CVRP	124.25	–	25.45	–	254.16	–	4.65	–
TDVRP	132.04	6.27%	24.19	–4.95%	247.91	–2.46%	4.07	–12.47%
TDVRP–PF	127.94	2.97%	23.83	–6.37%	243.20	–4.32%	4.06	–12.69%

Table 6
Solutions comparison between TDVRP and TDVRP–PF.

Models	Same routes	Same assignments, different routes	Different assignments
TDVRP vs. TDVRP–PF	5	17	8

Table 4 compares the results of the TDVRP–PF under different discretization levels based on the average of 30 instances. We observe that the results of the 30 min and 60 min discretization levels are the same. We also observe that, from 30 min to 10 min discretization level, the objective function value (cost) reduces only 0.07%, while the solution time increases 45.5%. When we further examine the solutions, the TDVRP–PF with 30 min and 10 min discretization levels result in the same routes in the solutions, which holds in all 30 instances. The reduction in cost with 10 min discretization level is due to the shorter time-dependent paths used along some arcs. This means that the marginal benefit of finer discretization is very small.

Considering reasonable solution time in the experiments, we conservatively select the 30 min discretization level to generate candidate time-dependent paths for the TDVRP–PF throughout the following experiments. Note that, the results are upper bounds of the optimal objective function values of the TDVRP–PF with the complete (optimal) set of candidate paths, and therefore are also lower bounds on the benefit of path flexibility in the time-dependent vehicle routing.

6.2.2. Path flexibility

Using the baseline instances, we compare the classical distance-minimizing vehicle routing problem (CVRP), the cost-minimizing time-dependent vehicle routing problem *without* path flexibility (TDVRP), and the cost-minimizing time-dependent vehicle routing problem *with* path flexibility (TDVRP–PF). We use the distance-minimizing path on each arc in the CVRP and use the time-minimizing path based on the average speed (of the day) on each arc in the TDVRP, which is what is usually done, implicitly, in the literature.

In Table 5, we provide the results on distance, fuel consumption, cost, and travel time, by evaluating the optimal solutions of the three VRP models. The results are the average of 30 baseline instances (R1–R30). We use the CVRP results as benchmarks and show the relative changes of the TDVRP and TDVRP–PF results.

Note that, due to the symmetry of the distance matrix in our setting, the opposite orientation of the optimal CVRP solution remains optimal. For example, routes 0–4–5–8–0 and 0–8–5–4–0 have the same distance. However, the corresponding fuel consumption, cost, and travel time may differ. To ensure consistency in comparison, we select the CVRP solution with the smallest cost (among different orientations of the CVRP optimal routes), shown in the “CVRP” row. We also include the CVRP results with the largest cost (in the “CVRP(+)” row) for comparison.

The TDVRP improves over the CVRP, but at the expense of a significant increase in travel distance (6.27%). By contrast, with a smaller increase in travel distance (2.97%), the TDVRP–PF improves similarly in travel time (12.69% vs. 12.47%), but more significantly in cost reduction (4.32% vs. 2.46%) and in fuel consumption saving (6.37% vs. 4.95%). These results clearly illustrate the benefit of path flexibility in the time-dependent vehicle routing.

In Table 6, we compare the solutions of the TDVRP and TDVRP–PF. In 5 out of the 30 instances, both the TDVRP and TDVRP–PF solutions have the same routes in their solutions. Note that “the same route” means the vehicle visits the same set of customers in the same order. In 17 instances, they have the same vehicle–customer assignments but different routes. In the remaining 8 instances, they have different vehicle–customer assignments.

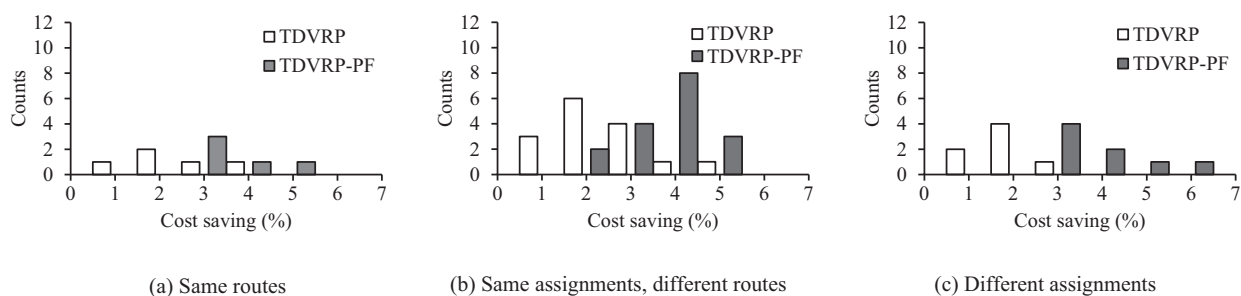


Fig. 9. Histograms of cost saving (vs. CVRP), grouped based on the solutions.

Table 7

Comparison results of instance R06.

Model	Distance		Fuel		Cost		Travel time	
	(km)	Diff. (%)	(liter)	Diff. (%)	(RMB)	Diff. (%)	(hour)	Diff. (%)
TDVRP	147.50	–	26.34	–	271.58	–	4.384	–
TDVRP-PF-r	140.37	–4.83%	26.16	–0.68%	267.05	–1.67%	4.495	2.63%
TDVRP-PF-a	139.38	–5.51%	26.11	–0.87%	266.18	–1.99%	4.502	2.76%
TDVRP-PF	140.16	–5.24%	25.85	–1.86%	264.50	–2.61%	4.303	–1.76%

In Fig. 9, we plot the instance count based on the cost saving of the TDVRP–PF and TDVRP, relative to the CVRP. Combining Table 6 and Fig. 9, we observe that, depending on the traffic condition of the road network, the TDVRP–PF may not simply change the path between two customer nodes within the same route, as compared to the TDVRP. Instead, by enabling path flexibility, the TDVRP–PF may result in structural changes, such as changes in routes or even vehicle–customer assignments, which enables more significant cost saving.

In what follows, as an illustration, we use a representative instance (R06), where the TDVRP and TDVRP–PF result in solutions with different vehicle–customer assignments, to illustrate the impact of path flexibility (similar observations can be made in other instances as well).

In Table 7, besides the results of the TDVRP and TDVRP–PF, we include two variants. In the TDVRP–PF-r, we fixed the routes as in the TDVRP solution, but allow for path flexibility, while in the TDVRP–PF-a, we fixed the vehicle–customer assignments as in the TDVRP solution, but allow for route changes and path flexibility.

We can observe in Table 7 that, path flexibility alone (TDVRP–PF-r) achieves the reduction of 4.83% in travel distance, 0.68% in fuel consumption, and 1.67% in cost, at the expense of 2.63% increase in travel time, compared to the TDVRP. Enabling route changes and path flexibility (TDVRP–PF-a) achieves further improvements in travel distance, fuel consumption, and cost, at the expense of longer travel time. The TDVRP–PF even changes the vehicle–customer assignments and achieves more significant cost saving (2.61%) with decrease in travel time (1.76%).

In Fig. 10, we plot the solutions of the TDVRP, TDVRP–PF-r, TDVRP–PF-a, and TDVRP–PF. In Fig. 11, we show the travel time distributions of the vehicles on different road categories.

By definition, the TDVRP and TDVRP–PF-r solutions have the same routes, which are route 1, (0-1-2-3-4-7-5-6-8-9-10-13-0), and route 2, (0-20-19-18-17-11-12-16-15-14-0). As shown in Fig. 10(a) and (b), among the 22 arcs, the TDVRP–PF-r results in the same paths as the TDVRP on 10 arcs and different paths on the other 12 arcs, for example, arcs (7-8-9) and (10-13) in route 1 and arcs (12-16) and (15-14) in route 2. As shown in Fig. 11(a) and (b), in the TDVRP–PF-r solution, vehicles travel significantly more on the residential roads during the peak hours (48.56%), comparing to that in the TDVRP solution (25.03%). Intuitively, TDVRP–PF-r solution trades travel time for lower total cost.

In the TDVRP–PF-a solution (Fig. 11(c)), route 2 remains the same as route 2 in the TDVRP and TDVRP–PF-r solutions, while route 1, (0-1-2-3-4-5-6-7-8-9-10-13-0), changes the visiting order of customer 7. The TDVRP–PF-a solution further increases the proportion of travel time on the residential roads during the peak hours to 51.49% (Fig. 11(c)), resulting slightly more reduction in cost than TDVRP–PF-r (1.99% vs. 1.67%), as shown in Table 7.

The TDVRP–PF solution changes the vehicle–customer assignments and moves customers 11 and 12 from route 2 to route 1 (Fig. 10(d)). The resulting routes are route 1, (0-1-2-3-4-5-6-7-8-9-10-11-12-13-0), and route 2, (0-14-15-16-17-18-19-20-0). Note also that the sequence of customer visits reverses in route 2.

Through the structural change in vehicle–customer assignments, customer visit sequences, and paths, the two vehicles in the TDVRP–PF solution spend significantly less time during the peak hours and more time during the off-peak hours (Fig. 11(d)). By avoiding traveling during the peak hours and congested roads, the TDVRP–PF achieves 2.61% of cost saving, comparing to the TDVRP.

In Table 8, we provide more detailed comparison between the TDVRP and TDVRP–PF solutions. For the convenience in comparison, we group the arcs into route segments. Note also that we reposition the route segments of route 2 in the

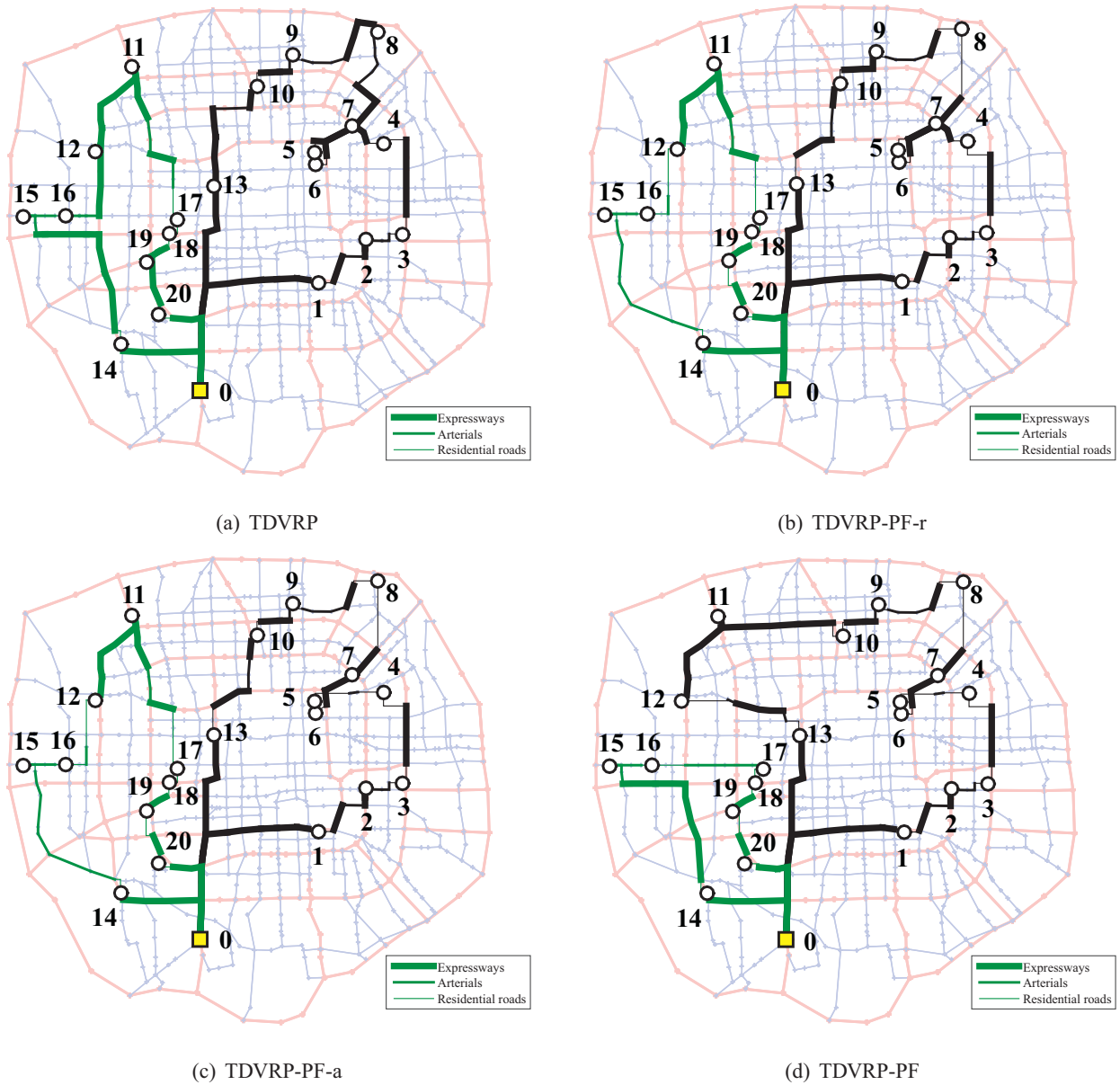


Fig. 10. Optimal solutions of instance R06.

TDVRP solution (e.g., (16–15–14–0) are the last arcs traveled by the vehicle) in order to make side-by-side comparison to the TDVRP–PF solution.

By moving customers 11 and 12 from between customers 17 and 16 in route 2 to route 1 and inserting them between customers 10 and 13, the route segments (10–11–12–13) and (16–17) in the TDVRP–PF solution save 4.16 RMB (32.27–28.11) in cost, compared to the route segments (10–13) and (17–11–12–16) in the TDVRP solution. We also observe that customers 11 and 12 are visited during the peak hours in the TDVRP solution yet are visited after the peak hours in the TDVRP–PF solution. Further, by changing the visiting order of customer 7, the route segment (4–5–6–7–8) in the TDVRP–PF solution achieves 2.01 RMB saving in cost, compared to the route segment (4–7–5–6–8) in the TDVRP solution.

Illustrated by the analysis of instance R06, compared to the TDVRP solution, the vehicles in the TDVRP–PF solution travel less in distance and in time, resulting in significant reduction in fuel consumption and saving in cost, achieved by the structural change in vehicle–customer assignments together with path flexibility.

In 25 out of the 30 baseline instances, the TDVRP–PF solutions change routes or even vehicle–customer assignments comparing to the TDVRP solutions, which indicates the difference in the solution structures between the TDVRP and TDVRP–PF. When these structural changes happen, more significant cost saving is achieved, as compared to the solutions with the same routes.

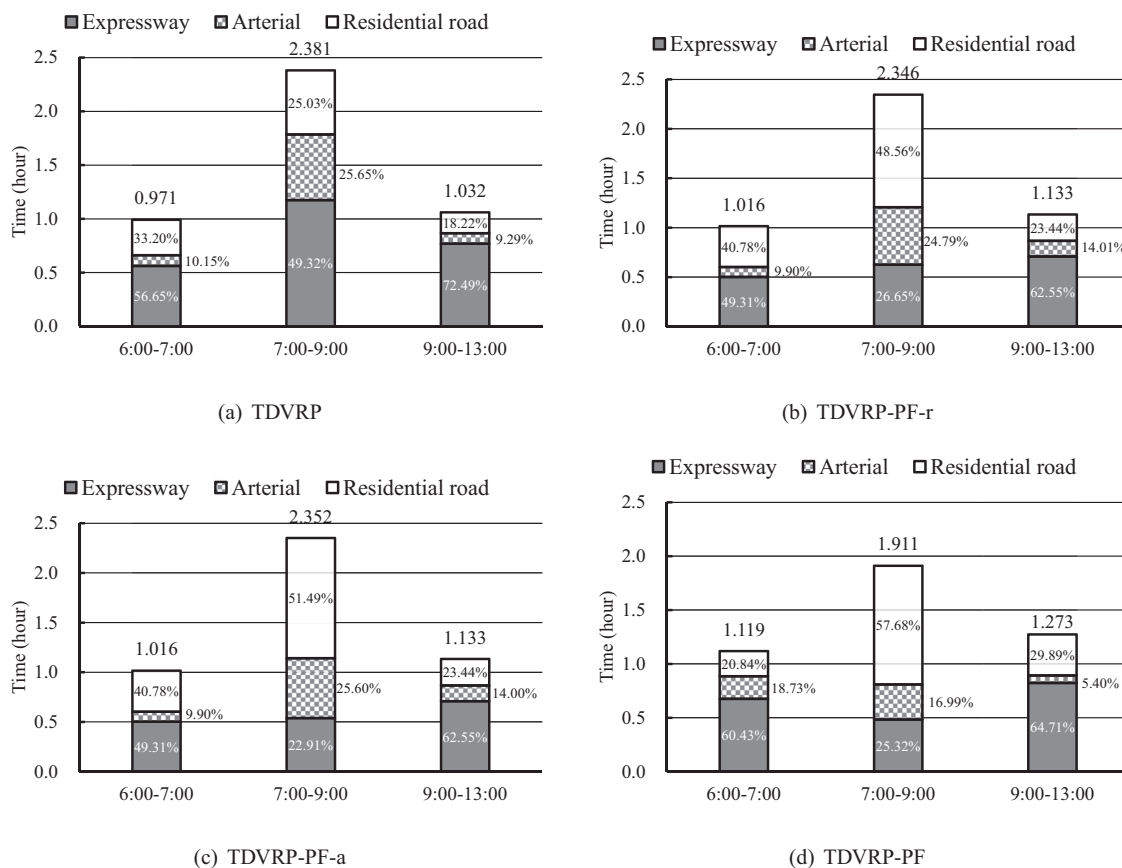


Fig. 11. Travel time distribution of instance R06.

Table 8
Details of R06 solutions.

TDVRP					TDVRP-PF						
Arcs	Distance (km)	Fuel (liter)	Cost (RMB)	Depart. time	Arcs	Distance (km)	Fuel (liter)	Cost (RMB)	Depart. time	Diff. cost (RMB)	Cost saving (%)
0-1-2-3-4	29.26	5.47	55.81	6:00	0-1-2-3-4	28.81	5.58	56.44	6:00	0.63	-8.90%
4-7-5-6-8	18.96	3.57	36.38	7:28	4-5-6-7-8	16.50	3.46	34.37	7:28	-2.01	28.39%
8-9-10	11.72	2.11	21.74	7:46	8-9-10	10.81	2.12	21.41	7:45	-0.33	4.66%
10-13	8.98	1.52	15.88	9:28	10-11-12-13	24.49	4.23	43.99	9:28	28.11	-397.03%
13-0	13.63	2.22	23.47	9:54	13-0	13.63	2.22	23.47	10:42	0	0.00%
Route 1	82.55	14.85	152.85		Route 1	94.23	17.62	179.68		26.41	-373.02%
16-15-14-0	23.52	3.85	40.55	8:24	0-14-15-16	23.52	3.95	41.40	6:00	0.85	-12.01%
17-11-12-16	24.99	4.60	47.06	7:06	16-17	6.93	1.50	14.79	7:04	-32.27	455.79%
0-20-19-18-17	16.43	2.99	30.70	6:00	17-18-19-20-0	15.48	2.78	28.63	7:31	-2.07	29.24%
Route 2	64.94	11.49	118.73		Route 2	45.93	8.23	84.82		-33.49	473.02%
Total	147.50	26.34	271.58		Total	140.16	25.85	264.50		-7.08	100.00%

6.2.3. Time flexibility

We consider three levels of time flexibility: TF-C, which allows up to 10 min post-service waiting time at each customer site, TF-D, which allows vehicles to postpone their departure from the depot up to one hour, and TF-CD, which is the combination of TF-C and TF-D.

The comparison in cost is shown in Table 9. Note that the row “Saving of PF” is the cost saving of the TDVRP-PF compared to the TDVRP, given the same level of time flexibility, and the columns “Saving of TF” are the cost saving of time

Table 9

Path flexibility vs. time flexibility: cost.

	No TF	TF-C		TF-D		TF-CD	
	Cost (RMB)	Cost (RMB)	Saving of TF (%)	Cost (RMB)	Saving of TF (%)	Cost (RMB)	Saving of TF (%)
TDVRP	247.91	247.91	0.00%	247.62	0.12%	247.62	0.12%
TDVRP-PF	243.20	243.15	0.02%	242.83	0.15%	242.80	0.16%
Saving of PF (%)	1.90%	1.92%		1.93%		1.95%	

Table 10

Path flexibility vs. time flexibility: travel time.

	No TF	TF-C		TF-D		TF-CD	
	Time (hour)	Time (hour)	Saving of TF (%)	Time (hour)	Saving of TF (%)	Time (hour)	Saving of TF (%)
TDVRP	4.07	4.07	0.00%	4.06	0.25%	4.06	0.25%
TDVRP-PF	4.06	4.06	0.00%	4.05	0.25%	4.05	0.25%
Saving of PF (%)	0.25%	0.25%		0.25%		0.25%	

Table 11

Path flexibility vs. time flexibility: waiting time.

	TF-C	TF-D	TF-CD		Total (hour)
	Waiting-C (hour)	Waiting-D (hour)	Waiting-C (hour)	Waiting-D (hour)	
TDVRP	0.00	0.37	0.00	0.37	0.37
TDVRP-PF	0.01	0.55	0.04	0.51	0.55

Table 12

Comparison among CVRP, TDVRP, and TDVRP-PF, by customer network topology.

		Distance		Fuel		Cost		Travel time	
		(km)	Diff. (%)	(liter)	Diff. (%)	(RMB)	Diff. (%)	(hour)	Diff. (%)
R	CVRP	124.25		25.45		254.16		4.65	
	TDVRP	132.04	6.27%	24.19	−4.95%	247.91	−2.46%	4.07	−12.47%
	TDVRP-PF	127.94	2.97%	23.83	−6.37%	243.20	−4.32%	4.06	−12.69%
RC	CVRP	119.37		24.17		241.95		4.39	
	TDVRP	126.73	6.17%	23.13	−4.30%	237.27	−1.93%	3.89	−11.39%
	TDVRP-PF	122.96	3.00%	22.76	−5.83%	232.66	−3.84%	3.87	−11.85%
C	CVRP	119.67		23.99		240.76		4.32	
	TDVRP	126.52	5.72%	22.94	−4.37%	235.68	−2.11%	3.83	−11.34%
	TDVRP-PF	122.67	2.51%	22.66	−5.54%	231.77	−3.73%	3.84	−11.11%

flexibility (TF-C, TF-D, and TF-CD) compared to the same model without time flexibility (No TF). We observe that time flexibility results in slight improvements, ranging from 0.00% to 0.16%, while path flexibility achieves consistent improvements, regardless of the levels of time flexibility, ranging from 1.90% to 1.95%.

We further compare the travel time and waiting time in Tables 10 and 11. We observe that the saving in travel time enabled by time flexibility is very limited. Although allowing time flexibility at the depot helps to save 0.01 h travel time on average, it costs the vehicles to wait 0.37 h in the TDVRP solution and 0.55 h in the TDVRP-PF solution. Note that the time flexibility (at the depot or at each customer site) within its upper limit is considered “free” here, which is rarely the case in practice, where the driver may need to pay for parking or to consider customer time windows. The results suggest that, although waiting (at the depot or at customer sites) to avoid the traffic congestion seems to be an intuitive and practical way to save the delivery cost, the benefit of waiting is not significant comparing to the benefit of flexibly selecting the travel paths.

6.2.4. Customer network topology

In Table 12, we compare the CVRP, TDVRP, and TDVRP-PF by customer network topology. We observe that the benefit of path flexibility becomes less significant when the customers become more clustered.

Table 13

Solution between TDVRP and TDVRP-PF: customer network topology.

	Same routes	Same assignments, different routes	Different assignments
R	5	17	8
RC	7	15	8
C	10	18	2

Table 14

Speed pattern under stochastic traffic conditions.

	Expressways (+/0/–)	Arterials (+/0/–)	Residential roads
06:00–07:00	56/48/40 km/h	40/32/24 km/h	24 km/h
07:00–09:00	48/36/24 km/h	32/24/16 km/h	24 km/h
09:00–13:00	50/41/32 km/h	36/28/20 km/h	24 km/h

Table 15

Evaluated results under stochastic traffic conditions.

	Deterministic		Route-Path		Stochastic	
	Cost (RMB)	Diff. (%)	Cost (RMB)	Diff. (%)	Cost (RMB)	Diff. (%)
TDVRP	165.29				165.29	0.00%
TDVRP-PF	163.71	–0.96%	161.75	–2.14%	161.73	–2.15%

We further compare the TDVRP and TDVRP-PF solutions in Table 13 and observe that, when the customers are more clustered, the structure of the TDVRP and TDVRP-PF solutions becomes more “similar.” That is, there are fewer solutions with different customer-vehicle assignments and more solutions with the same routes. This is reasonable, as it may become more efficient and economical for the vehicle to serve nearby customers in clusters, when they are close to each other.

6.2.5. Stochastic TDVRP-PF

In this subsection, we study the impact of path flexibility under stochastic traffic conditions. However, we do not intend to solve large scale stochastic and dynamic problems. Rather, inspired by Lium et al. (2009), we study a simplified two-stage stochastic problem (“a kind of stochastic equilibrium model”), as described in Section 5.2, to obtain key insights on the interaction between path flexibility and stochasticity.

Based on the traffic data, the traffic conditions on the residential roads are relatively stable during the day. Therefore, we only consider major speed situations on expressways and arterials. Denote ‘+’ as the less congested situation, ‘0’ as the normal situation, and ‘–’ as the more congested situation. The probability of the normal situation is 40%, and the probability of the other two situations is 30%. Thus, there are in total nine scenarios, considering the combinations of three situations of each of the two road types (expressways and arterials). The speed pattern under stochastic traffic conditions is shown in Table 14.

We solve 10 R instances each with 10 customers served by one vehicle. In each instance, the 10 customers are randomly selected from the corresponding 20-customer R instance under deterministic traffic conditions.

We compare the results of three models: the deterministic model that solves the TDVRP-PF for both routing and path decisions under the expected traffic condition, the stochastic model that solves the two-stage stochastic program TDVRP-PF^S, and a Route-first-Path-second (Route-Path) approximation method. In the Route-Path approximation method, we obtain the routing decisions via solving the deterministic TDVRP-PF, and then allow for path decisions to adapt to realized traffic scenarios. That is, the Route-Path approximation method decides upon the vehicle-customer assignments and vehicle routes first via solving the deterministic TDVRP-PF under the expected traffic condition, rather than the two-stage stochastic program. Then, with the vehicle-customer assignments and vehicle routes fixed, the Route-Path approximation method allows for path selection flexibility in response to the realized traffic scenario. This is inspired by Bai et al. (2014), where they propose similar decomposition-like approximation methods to solve a two-stage stochastic service network design problem with rerouting.

The solutions are then evaluated under the traffic scenarios and the average results of the 10 R instances are shown in Table 15. The deterministic TDVRP and stochastic TDVRP (or TDVRP^S) result in exactly the same solutions, due to the lack of recourse actions. On the other hand, the stochastic TDVRP-PF (or TDVRP-PF^S) increases the cost saving from 0.96% to 2.15%. Note that the “Diff” columns show the percentage differences in the evaluated objective function values relative to the TDVRP.

An important finding is that the Route-Path approximation method generates good solutions. Among the 10 instances, the Route-Path solution and the optimal solution of the stochastic TDVRP-PF obtain the same routes in 9 instances. In the only

instance where the two solutions are different, only the visit sequence of two consecutively visited customers is swapped, resulting in 0.11% increase in the evaluated cost. Consequently, the average evaluated cost of the Route-Path approximation method and the stochastic model are very close (only differ in 0.01%).

This observation suggests that the routing decision of the TDVRP–PF is somehow robust against traffic conditions and the path selection decisions serve as a natural recourse action in coping with different traffic conditions. In practice, logistics planners can decide the routes based on the expected traffic conditions, and drivers can select paths depending on real-time traffic conditions. Although this seems intuitive in practice, it has not been implemented and analyzed quantitatively.

On the other hand, as the Route-Path approximation method decomposes the two-stage stochastic program into separate master problem and subproblems, its solution time is significantly shorter (on average 23.93 s) than that of the TDVRP–PF⁵ (on average 110.75 h).

The above observations are encouraging. While we admit the computational challenges in solving problems that deal with path flexibility, time-dependency, and stochasticity, the satisfactory solution quality of the Route-Path approximation method provides a hope to cope with the challenges in practice.

7. Conclusions

In this paper, we emphasize the importance of explicitly considering two graphs in time-dependent vehicle routing problems, that is, the *customer graph* (with customer locations as nodes in the network) and the *geographical graph* (with inter-sections as nodes in the road network). Thus, each arc in the customer graph corresponds to multiple paths in the geographical graph. We explicitly consider the path selection in the road network as an integrated decision in the vehicle routing problem. The corresponding problem is a time-dependent vehicle routing problem with path flexibility (TDVRP–PF). We formulate the TDVRP–PF models both under deterministic and stochastic traffic conditions.

The experimental results demonstrate the benefits of path flexibility. Compared to the TDVRP *without* path flexibility, the TDVRP–PF results in significant savings in terms of cost and fuel consumption, by explicitly considering path flexibility in routing and scheduling. The role of time flexibility (i.e., flexible departure time at the depot and post-service waiting time at customer sites) appears to be less significant. Moreover, the results and insights obtained are robust on a wide range of instances (considering customer topology and the number of alternative paths). We also observe and analyze the structural difference between the TDVRP and TDVRP–PF solutions. Combining path flexibility and time-dependent travel time seems to be a good representation of a wide range of stochasticity and dynamics in the travel time, and path flexibility serves as a natural recourse under stochastic traffic conditions. Exploiting this observation, we employ a Route-Path approximation method, which obtains the routing decision based on expected traffic conditions and the en route path selection decision depending upon the real-time traffic conditions. This approximation method obtains good solution quality in shorter solution time and is easy to implement in practice.

Finally, this is exploratory research on incorporating path flexibility in the vehicle routing problems. Further research is needed. For example, the impact of path flexibility in other variants of the vehicle routing problems, such as the VRP with time windows, needs to be further studied. Clearly, the introduction of path flexibility in the vehicle routing problem significantly increases the problem size. It is important to study and develop efficient algorithms to solve large scale instances, especially under stochastic traffic conditions.

Acknowledgment

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Appendix A. CMEM model

In the Comprehensive Modal Emission Model (CMEM, [Barth et al., 2005](#) and [Barth and Boriboonsomsin, 2009](#)), for a vehicle traveling a distance d at a constant speed v in duration t ($t = d/v$) and carrying a load f , the fuel consumption $F(t, v, f)$ is calculated as

$$F(t, v, f) = \lambda k N_e V t \quad (\text{A.1})$$

$$+ \lambda \gamma \beta v^3 t \quad (\text{A.2})$$

$$+ \lambda \gamma \alpha d (\mu + f), \quad (\text{A.3})$$

where μ is the vehicle curb-weight, $\lambda = \varepsilon / (\kappa \psi)$, $\gamma = 1 / (1000 \epsilon \varpi)$, $\alpha = g \sin(\phi) + g_C \cos(\phi)$, and $\beta = 0.5 C_d A_f \rho$. In [Table A.16](#), we describe the parameters and the corresponding values based on a typical light-duty vehicle (vehicle type:

Table A.16
Parameters in the CMEM model.

Type	Notation	Description	Value
Vehicle-dependent	k	Engine friction factor (kJ/rev/liter)	0.20
	N_e	Engine speed (rev/s)	38.33
	V	Engine displacement (liter)	4.70
	A_f	Frontal surface area (m ²)	5.03
	μ	Curb-weight (kg)	3850
	Q	Vehicle capacity (kg)	4460
	ε	Vehicle drive train efficiency	0.4
	ω	Efficiency parameter for diesel engines	0.9
Road-dependent	ϕ	Road angle	0
	C_r	Coefficient of rolling resistance	0.01
Other constants	ξ	Fuel-to-air mass ratio	1
	κ	Heating value of a typical diesel fuel (kJ/g)	44
	ψ	Conversion factor (g/liter)	737
	ρ	Air density (kg/m ³)	1.2041
	g	Gravitational constant (m/s ²)	9.81
	C_d	Coefficient of aerodynamic drag	0.7

JAC HFC1082KD) in China and we adapt the value of other constants and road-dependent parameters from Franceschetti et al. (2013).

The CMEM model consists of three modules: the engine module (A.1), the speed module (A.2), and the weight module (A.3). For simplicity, We rewrite the fuel consumption in Eq. (1) with parameters ψ_e , ψ_s and ψ_w representing the coefficients of these three modules. Clearly, the fuel consumption rate depends on both the speed and load of the vehicle. Notice also that the speed and load related fuel consumptions are separable and additive in the CMEM model.

Appendix B. Traffic speed

We construct the traffic speed mainly based on official sources and partially based on the educational guess due to the inaccessibility of data.

We collected the speed data from the Beijing Municipal Commission of Transport (<http://www.bjjtw.gov.cn/bmfwjtzs/>) from October 14th to 18th (Monday to Friday), 2013. The average speeds are shown in Table B.17.

As reported in the government website (Beijing Government, 2015), the speed of expressways in morning peak hours in 2013 was 36.7 km/h and the speed of arterials in morning peak hours in 2013 was 22.9 km/h. We set the average speed of expressways in the morning peak hours equal to 36.0 km/h and the average speed of arterials in the morning peak hours equal to 24.0 km/h. We estimate the average speeds of expressways and arterials in off-peak hours according to the speed of entire road network in Table B.17. That is,

- Average speed of expressways in 6:00–7:00 = $36.0/28.9 \times 38.7 = 48$ km/h
- Average speed of expressways in 9:00–13:00 = $36.0/28.9 \times 33.3 = 41$ km/h
- Average speed of arterials in 6:00–7:00 = $24.0/28.9 \times 38.7 = 32$ km/h
- Average speed of arterials in 9:00–13:00 = $24.0/28.9 \times 33.3 = 28$ km/h

Meanwhile, we adjust the speeds of arterials in different areas (e.g., inside the 2nd ring) to show the spatial difference of congestion. For example, in morning peak hours, the ratio of the speed of arterials inside 2nd ring to the speed of arterials between 2nd ring to 3rd ring is 23.0/27.0. To reflect the spatial difference of congestion on expressways, we lower the speed of part of the ring roads, which are notorious for severe congestion during peak hours. The detailed speed setting of expressways and arterials is in Table B.18.

Table B.17
Speed of Beijing road network.

	Speed (km/h)		
	6:00–7:00	7:00–9:00	9:00–13:00
Entire road network	38.7	28.9	33.3
Inside 2nd ring	32.9	23.0	26.2
2nd ring to 3rd ring	36.6	27.0	30.4
3rd ring to 4th ring	43.5	32.3	36.7
4th ring to 5th ring	43.6	33.4	38.2

Table B.18
Speed setting of the numerical experiment.

		# road segments	06:00–07:00	07:00–09:00	09:00–13:00
Expressways	East 2nd and 3rd rings	28	36	20	30
	West 2nd ring	20	36	18	30
	North 2nd ring	10	40	20	36
	North 3rd and 4th rings	38	45	20	36
	Others	390	51	38	44
		Average	48	36	41
		# road segments	06:00–07:00	07:00–09:00	09:00–13:00
Arterials	Inside 2nd ring	216	28	19	23
	2nd to 3rd ring	236	29	22	25
	3rd to 4th ring	382	33	26	30
	4th to 5th ring	490	34	28	31
		Average	32	24	28

Appendix C. Construction of cluster instances

We construct the C (clustered) instances based on the procedure described in Algorithm 2. In the numerical experiments, we set the minimum number of clusters $\underline{C} = 2$, the maximum number of clusters $\bar{C} = 4$, and the minimum distance between cluster centers $D = 10$ km.

To generate the RC instances, we generate 2 clusters and select 10 customers based on Algorithm 2. The other 10 customers are randomly selected from the customer set.

The customer locations of the first ten of the R, RC, and C instances are shown in Figs. C.12, C.13, and C.14, respectively.

Algorithm 2 Construction of the C (clustered) instances.

- Step 1. Generate the number of clusters C from the discrete uniform distribution $[\underline{C}, \bar{C}]$.
- Step 2. Randomly select customer nodes $n_1, \dots, n_C$ as the centers of these clusters, such that $d_{n_i, n_j} \geq D$, where $1 \leq i < j \leq C$. Update the candidate customer set $N^c = N^c \setminus \{n_1, \dots, n_C\}$.
- Step 3. For every candidate customer $n \in N^c$, assign a value $v_n = \max_{1 \leq i \leq C} \{1/d_{n, n_i}\}$.
- Step 4. Calculate the probability that candidate customer $n \in N^c$ is chosen, $p_n = v_n / (\sum_{n \in N^c} v_n)$.
- Step 5. Randomly select a customer $n' \in N^c$ based on the probabilities of all candidate customers. Update the candidate customer set $N^c = N^c \setminus \{n'\}$.
- Step 6. If N customers are chosen, stop. Otherwise, go back to Step 4.

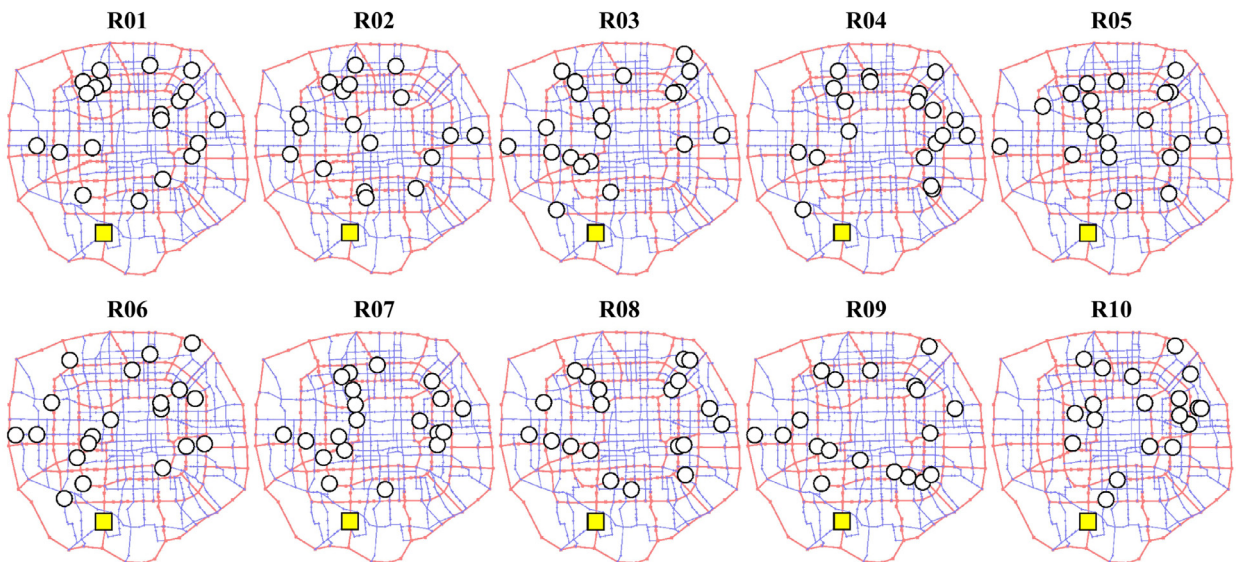


Fig. C.12. Customer locations of the R instances.

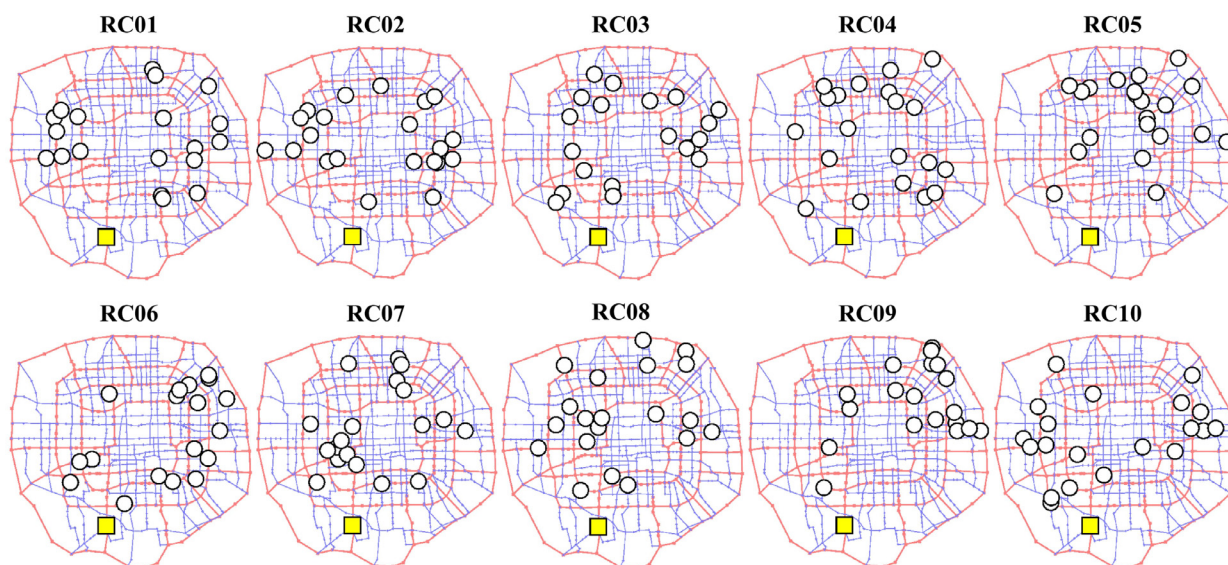


Fig. C.13. Customer locations of the RC instances.

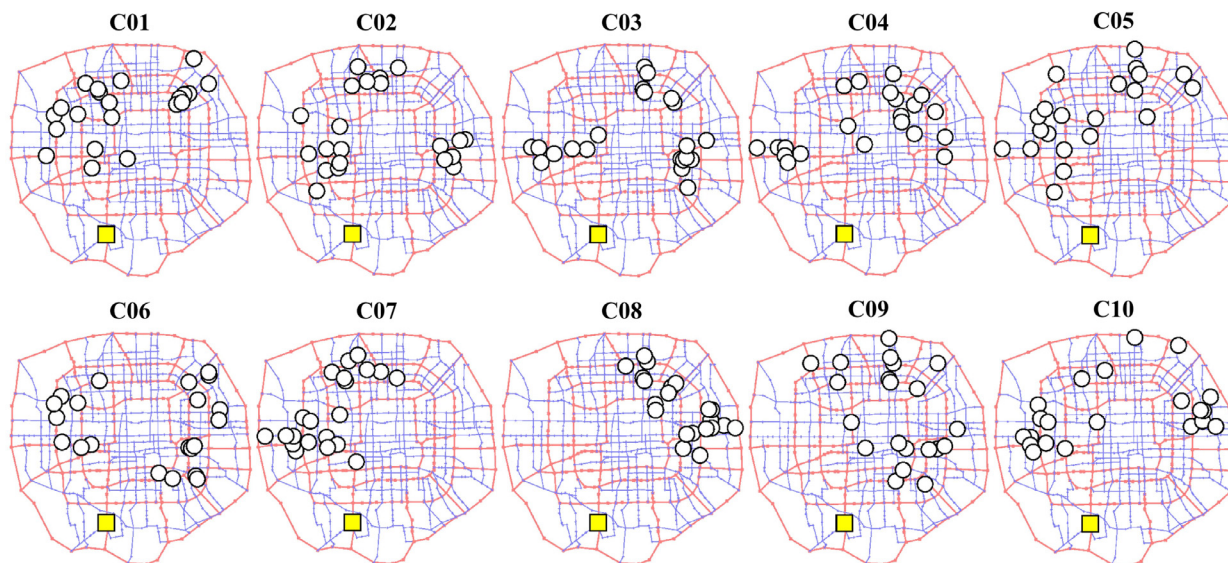


Fig. C.14. Customer locations of the C instances.

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